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PROPERTIES OF WATER / STEAM			
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Appendix E

Properties of water and steam above 100°C

Contents:

1 Page (steam table)

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State of saturation of water at vacuum

Temperature:	10	mbar: 9,1
	20	17,4
	30	32,0
	40	55,0
	50	94,0

Steam table (state of saturation of water at excess pressure)

Temperature	Steam excess pressure	Spec. Volume	Water expansion	Density
°C	bar	m ³ /kg	%	kg/Ltr.
100	0,03	—	4,35	0,0006
125	1,4	—	6,50	0,0013
140	2,7	—	7,98	0,0020
150	3,9	—	9,06	0,0025
160	5,3	—	10,21	0,0033
170	7,1	—	11,44	0,0041
180	9,2	—	12,75	0,0052
190	11,8	—	14,15	0,0064
200	14,9	0,128	15,65	0,0079
210	18,5	0,105	17,26	0,0096
225	25,0	0,078	19,91	0,0128
240	33,1	0,059	22,91	0,0168
250	39,6	0,050	25,12	0,0200
260	46,9	0,041	27,55	0,0237
270	55,1	0,035	30,23	0,0281
275	59,7	0,032	31,69	0,0306
280	64,5	0,030	33,21	0,0332
285	69,5	0,027	34,84	0,0361
290	74,9	0,025	36,55	0,0392
295	80,6	0,024	38,37	0,0426
300	86,6	0,022	40,36	0,0462

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Appendix F

Safety valve(s), rupture disc and contact manometer

Contents:

Drawing no. 152466 A

Mounting of safety valve

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1.0 SAFETY VALVES

1.1 FUNCTION

Every pressurised vessel has to be equipped with a safety valve to provide, that the system will run at a higher pressure than the max. allowable pressure p_0 .

1.2 TECHNICAL DATA

Type test approved as full lift safety valve for steam/gases

Appr. No.: TÜV-SV 98-934 D/G/F

Coefficient of discharge for steam/gases $\alpha_d=0,8$

Body material 1.0619/ WCB

Bonnet material 1.0619/ WCB

Flange connection

Inlet: DN 25 PN 160, DIN 2638

Outlet: DN 50 PN 40, DIN 2635

1.3 INSTALLATION

General notes

The quality of the sealing surfaces is endangered by the ingress of foreign materials. Impurities must be prevented from entering the valve during transportation, installation and operation.

When installing safety valves with threaded connections, use metal seal washers only. Seal materials such as hemp or PTFE tape should not be used as this type of material can break off and enter the valve causing it to leak.

If valves with open bonnets and / or levers are to be painted, care must be taken to protect the sliding parts; otherwise the correct operation of the safety valves may be affected.

Transport Protection

All protective devices at transport and handling have to be removed before mounting the safety valve.

The lever of spring loaded safety valves is secured with a wire to the bonnet subject to the pressure setting. This wire should be removed only after installation. For testing the set pressure, or tightness of the safety valve, remove this wire and take care that the lever is not engaged with the spindle cap.

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Valves supplied with lever and weight are packed with a wooden wedge to secure the lever and prevent movement of the disc. This packing precaution should only be removed after installation.

Assembly

All drainage pipes that have been connected must be free of restriction and the ends easily observed. Care must be taken when discharging any drainage to prevent injury to personnel (i.e. on steam lines connect a suitable steam trap).

Attention! When valves are supplied with a drain hole that is not going to be used the plastic plug fitted in the hole must be removed and replaced by a mental plug.

The discharge piping on steam lines should be installed in an ascending direction, and on liquid lines in a descending direction. Spring loaded safety valves should be installed with the bonnet VERTICALLY UPWARDS.

Safety valves should be mounted in such a way that no inadmissible static, dynamic or thermal loads can be transmitted to the valve due to up or downstream pipe work. If necessary expansion devices should be utilised. Additional stresses by incorrect mounting should be avoided. Freezing and sticking of fluid or condensate in the valve body should be avoided by suitable actions.

To prevent the ingress of dirt and other impurities from entering the safety valve, drainage of the discharge pipe and safety valve must be implemented via the discharge pipe. Therefore, safety valves are not provided with drain holes.

According to technical regulations, a drain hole of sufficient size must be incorporated at the lowest point of pipe work. The discharge pipe must slope in a downward direction and be fitted with a drainpipe (which should be DN 25 minimum for discharge pipes DN 40 and larger) before any bends are connected.

Set point adjustment

The pressure setting is wire-locked and sealed against unauthorised alterations. TÜV regulations, agreed by the manufacturer, require the fitting of a type test approval plate stating the correct valve data. The manufacturer can not be held responsible for any changes to set pressure or other alterations after despatch from the factory. In case of any necessary modifications, we recommend to return the valve to the supplier, an authorised repair shop or to do the modifications under the supervision of the TÜV or any other suitable inspecting authority.

Lifting device H3 (See figure 1 below)

- | | |
|---------|--|
| Step 1. | Remove shaft (45) |
| Step 2. | Pull lever (43) from lever cover |
| Step 3. | Loosen screw (84) |
| Step 4. | Unscrew and remove lever cover (41) |
| Step 5. | Slacken lock nut (19) |
| Step 6. | Turn adjusting screw (18) to the required set pressure |

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Note: Draw attention to the admissible pressure range of the spring!

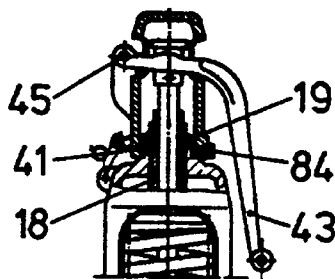


Figure 1 Lifting device H3

Turning the adjusting screw (18) in a clockwise direction increases the spring tension, thus obtaining a higher set point pressure. Turning the adjusting screw anticlockwise reduces the spring tension, thus obtaining a lower set point pressure.

Upon completion of adjustment, reassembly is in reverse order of steps 1 – 6 above, this locks out the set pressure.

1.4 INSPECTION AND TESTING

The safety valves are tested, adjusted to the required set pressure and sealed (see also DIN 3320). The set pressure and the LESER sign is marked on the edge of the outlet flange.

If the TÜV or any other licensed authority inspects the valves, their signs are also hard stamped on the edge of the flange or on the valve body.

The valve should be closed by sudden release of the lever. If this procedure does not stop the leakage, it is likely that the valve seat is damaged. Repairs should be carried out at GEKAKONUS GMBH 's workshop or by an authorised person.

It is recommended to lift the valve lever manually from time to time in order to prevent accumulation of deposits that will affect valve operation (testing of safety valves is required for plants operating with demineralised water and for hot water generators at intervals of a least six months).

It is important to ensure that the lifting lever does not engage the spindle cap after the lift operation. The lever must be deflected towards the centre-line of the bonnet until the lifting fork is disengaged.

1.5 OPERATION AND SAFETY

The working pressure of the plant should be at least 5% below the reseating pressure of the valve to enable a correct re-seat. The valve should be closed by sudden release of the lever. If

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this procedure does not stop the leakage, probably the valve seats are damaged. Repairs should be carried out at our workshop.

1.6 MAINTENANCE

It is recommended and in Germany required by rules, to lift manually the valves lifting lever from time to time in order to prevent accumulation of deposits that will affect the valve operation.

2.0 RUPTURE DISC

2.1 FUNCTION

A rupture disc has to be installed to provide leakages of the system. as the safety valve begins to open in a range of abt. 3-5 % below the set pressure.

2.2 TECHNICAL DATA

Type: RLS-90
Size: DN 25
Material: 1.4401

For more details see the "Burst Test Certificate".

2.3 INSTALLATION

The rupture disc will be installed boiler connection flange and the inlet connection of the safety valve, for details see drawing no. 152466 in this appendix.

2.4 INSPECTION AND TESTING

Check knife blades in outlet flange for sharpness. Dullness may affect rupture disc performance. If slightly dull or nicked, sharpen with mill file or stone. If nicks cannot be filed out, then replace flange. When penetration points are attached to the knife blades, then replace outlet flange. When penetration points are attached to the knife blades, they must be sharp and not bend.

2.5 OPERATION

During the normal operation the rupture disk is not damaged.

2.6 MAINTENANCE

Corrosion and service conditions may affect disk life, thus requiring periodic change. Regular disk inspection is advised every 3 month to insure that corrosion, creep, etc. are not a threat to predictable performance.

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3.0 CONTACT MANOMETER

3.1 FUNCTION

The contact manometer is used to monitor the pressure between the rupture disk and the safety valve.

3.2 TECHNICAL DATA

Name:	Contact manometer
Type:	237 001
Contact:	1 opener as magnet spring-contact
Range:	0-100 bar
Set point:	10 bar
Casing diameter:	100 mm
Indication accuracy:	class 1,0
Excess pressure safety:	max. 130 bar short- term
Contact load:	380 V - 30 W / 50 VA 380 V - 80 mA 220 V- 135 mA 110 V - 270 mA
Contact material:	Silfram W 30 = Silver + 30 % Wolfram
Ambient temperature:	-20....+140°C
Type of switch:	W1/101
Connection:	R 1/4", below

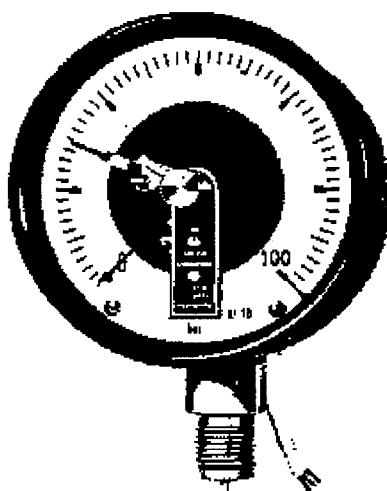


Figure 2 Contact manometer

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3.3 INSTALLATION

For installation details see drawing no. 152466 B in this appendix.

3.4 INSPECTION AND TESTING

The contact manometer could be tested via the testing cock at the front side see drawing no. 152466 B

3.5 OPERATION

Because of the rupture disc the manometer is not pressurised, the contact is closed. By auxiliary relays with indication lamps or auxiliary electric circuits, normal system operation could occur.

Activation of the rupture disc will mean that the full pressure is lying on the manometer, the contact will open and the related trouble indication lamp signal "excess pressure rupture disk". If this alarm is given the rupture disk is broken and has to be replaced.

3.6 MAINTENANCE

No special maintenance is required.

Beim Abblasen eines Sicherheitsventils entstehen Reaktionskräfte, die vom Ventil selbst, den sich anschließenden Leitungen und den Festpunkten aufgenommen werden müssen. Dabei ist zu beachten, daß keine unzulässigen statischen, dynamischen oder thermischen Beanspruchungen aus den zu- und abführenden Rohrleitungen auf das Ventil übertragen werden. Die zu-lassigen Beanspruchungen ergeben sich aus dem zulässigen Biegemoment.

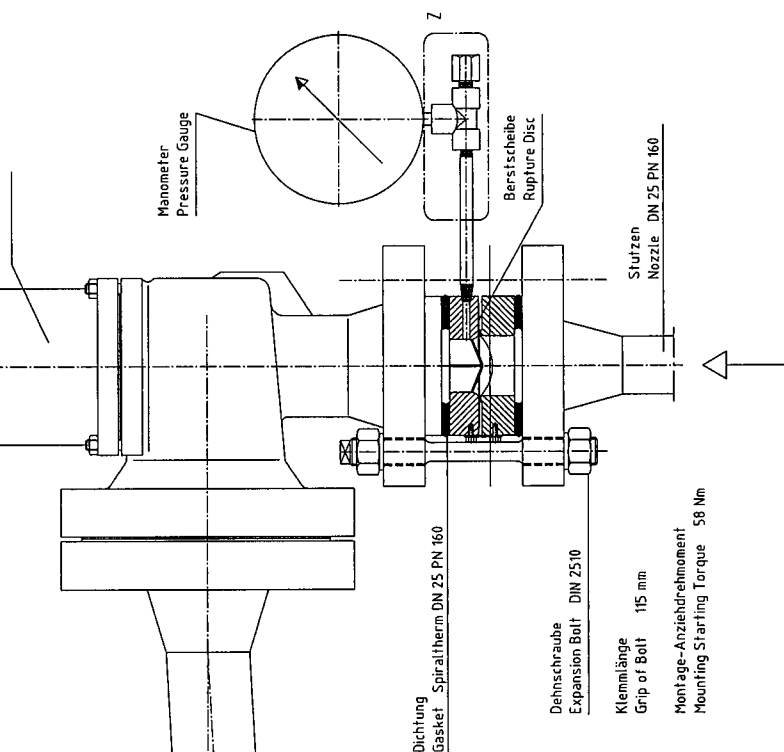
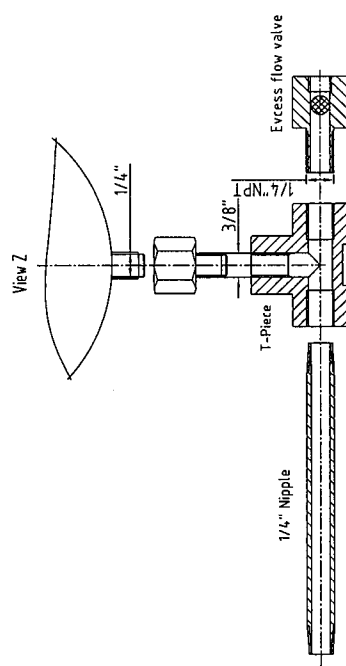
Um Schmutz und Fremdkörper aller Art von dem Sicherheitsventil fernzuhalten, muß eine Entwässerung der Ausblaseleitung unter dem Sicherheitsventil über die Ausblaseleitung erfolgen. Darum ist normalerweise keine Entwässerungsbohrung am Sicherheitsventil vorgesehen.

Technical drawing showing a cross-section of a drainage system. The drawing includes a pipe with a drainage outlet labeled "Entwässerung Drainage" and a drainage channel labeled "Gefälle Descent". The pipe is shown with a cross-section and a longitudinal section. The drainage outlet is a vertical pipe with a flange and a cap. The drainage channel is a horizontal pipe with a flange and a cap. The drawing is a technical illustration of a drainage system.

Repulsion powers develop at the blowing off of the safety valve, which have to be absorbed by the valve itself, the following pipes and the fix points. It has to be considered, that no admissible static, dynamic or thermic loads are transferred to the valve from the inlet or outlet pipes. The admissible loads follow from the admissible bending moments.

To keep away dirt and all kind of foreign bodies from the safety valve, the draining of the blow-off pipe and the safety valve must be done over the blow-off pipe. For that reason there is normally not a drainage boring at the safety valve.

Excess flow valve must be install in horizontal position.

[illegible]

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Appendix G

Installation instructions for NUK HP systems

Drawing no. 154246 C

Example for pipe work NUK-HP

Drawing no. 154546 B

Max. admissible forces

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1.0 General

It is the responsibility of the owner (client) of NUK HP systems to ensure that an acceptable standard of workmanship is achieved at the time of the site installation.

Due diligence must be given to the quality of site fabrication, particularly the welding, because it cannot be stressed enough that the operating system must be pressure tight to ensure reliable service.

1.1 Design

A suitable nationally or internationally recognised code of construction should be adopted for the design and construction requirements of high pressure steam and condensate pipe work.

Such a code could be; ASME/ANSI B3.1 – Power piping.

The guidelines listed below must also be incorporated into the final system design layout.

Steam and condensate lines in working order must be installed with a constant slope of at minimum 5% from the heat consumers to the NUK. (i.e. If a contra-slope should develop due to the thermal expansion of the pipe, this can result in low water problems and strong steam hammering.

At operational temperatures (300 °C) the steam and condensate lines expand by 3.8 mm per meter (at 10 m long pipe sections by 38 mm).

With regard to the change of length, the steam and condensate lines must be designed such that the as-built construction includes sufficient length for compensation. Fix- and sliding points for the rising reaction power must be appropriately calculated.

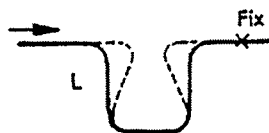
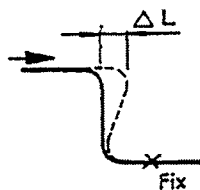
The laying of pipe-work must happen with adequate pre-stressing - i.e. half of the change of length must be installed in cold condition - opposite to the expansion direction.

Example of pipework compensation:

Expansion absorption: at L1 of 50 mm results a leg length L of:

- For a tube 42,4 x 2,9 = 5 m
- For a tube 60,3 x 3,2 = 6,5 m
- For a tube 76,1 x 3,6 = 8 m

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Pipework engineering has to be done by competent companies, the complete installation has to be accepted by the local authorities.

1.2 Materials of construction

Tube diameter:	< DN 65, St. 35.8, Quality I,	ASME, SA 192 or SA 220
	> DN 65, St. 35.8, Quality III	ASME, SA 192 or SA 220

Welding material autogenous: Material no. 1.6215

Welding material electro-arc welding: Gricon 8 A or similar

1.3 Inspection and testing

The non-destructive testing requirements stated in the relevant code of construction should be implemented on the pipework.

As a minimum this should include:

- 100% Visual examination on all weld connections
- 10% Radiographic examination on randomly selected (completed) butt welds
- 10% Examination of welding seams by dye penetration test

Upon completion, the system cleaning and pressure testing procedures stated in chapter 3 of this manual should be implemented and the appropriate certification issued (if satisfactory).

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The recommendation is based on following average operating conditions:

You must use the size of pipe dimension that you have calculated according to chapter 3 (page5/9). This required minimal pipe dimension is based on following average operating conditions:

Saturated steam

Operating pressure

Evaporating heat

Volume specific

Steam velocity

$p = 50 \text{ bar}$

$r = 390,0 \text{ kcal/kg}$

$v'' = 0,03942 \text{ m}^3/\text{kg}$

$< 10 \text{ m/s}$

Tubes acc. to DIN 17175 or 17155 seamless or welded					Tubes acc. to ASA B 36.10. Schd. no. 80 seamless or welded					
Boiler type	Tube *)	Tube dimension	Tube content	Tube weight insul. and filled with water	Tube *)	Tube dimension	Tube content	Tube weight insul. and filled with water	Insul thickness at 300°C tube temp.	Heat loss
NUK-HP	DN [mm]	$d_a \times s$ [mm]	[l/m]	[kg/m]	DN [inch]	$d_a \times s$ [mm]	[l/m]	[kg/m]	s [mm]	[W/m]
320	40	48,3x3,6	1,33	17	1 1/2"	48,3x5,08	1,14	18,5	70	87
435	40	48,3x3,6	1,33	17	1 1/2"	48,3x5,08	1,14	18,5	70	87
700	50	60,3x4,5	2,15	22	2,0"	60,3x5,54	1,9	23	80	92
930	50	60,3x4,5	2,15	22	2,0"	60,3x5,54	1,9	23	80	92
1250	65	76,1x5,0	3,43	27	2 1/2"	73,0x7,01	2,73	30	80	98
1500	65	76,1x5,0	3,43	27	2 1/2"	73,0x7,01	2,73	30	80	98
1745	80	88,9x6,3	4,57	32	3,0"	88,9x7,62	4,26	34	80	107
2100	100	114,3x8,0	7,59	42	4,0"	114,3x8,56	7,61	43,5	100	110
2325	100	114,3x8,0	7,59	42	4,0"	114,3x8,56	7,61	43,5	100	110
3500	100	114,3x8,0	7,59	42	4,0"	114,3x8,56	7,61	43,5	100	110
4200	125	139,7x10,0	11,71	66,0	5,0"	141,3x9,52	11,74	69,0	120	120
5350	150	168,3x12,5	16,12	96,0	6,0"	168,3x11,0	16,80	90,0	150	135

In case it should be not possible in practice, to install the calculated steam- and condensate line, please chose a larger tube diameter which shows a sufficient length for the real installation.

*) These tube dimensions are minimal tube dimension that must not be smaller

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Calculation sheet for the inside diameter of the steam- and condensate connection as well as length

Type:	NUK-HP			(1)
Operating pressure max.:			bar	(2)
Boiler contents:			litre	(3)
Steam- and condensate connection Tube dimension ($d_a \times s$):		x	mm	(4)

A.) Calculation of tubing length to be installed and diameter

Boiler contents (from level ∇^{---1} to ∇^{---2})			litre	(1)
Safety filling quantity = 20% of boiler contents (from level ∇^{---2} to ∇^{---3})			litre	(5)
Total filling quantity for start of plant (from level ∇^{---1} to ∇^{---3})	=		litre	(6)

For the calculation of tubing dimension to be installed ($\emptyset$ and length) between NUK-HP and the heat exchanger for heating of vegetable oil, the volume expansion of boiler water – relating to the total filling quantity for the start of plant (6) and the operating pressure max. has to be considered.

The volume expansion of boiler water is:

- at heating-up from 20°C / 1 bar to 287°C/ 70 bar excess pressure: 35% (7)
- at heating-up from 20°C/ 1 bar to 304°C/ 90 bar excess pressure: 42% (8)

The contents in the system at operating pressure max. is calculated from the total filling quantity for the start of plant (6) plus the respective volume expansion (7) resp. (8) in regard of (6).

Contents in the system at operating pressure max.			litre	(6)
Total filling quantity for the start of plant (level ∇^{---3})	=			
Volume expansion (7) resp. (8)	=		litre (7) resp. (8)	(8)
Contents in the system at operating pressure max.: (level ∇^{---4})	=		litre	(9)

The contents in the system at operating pressure max. is of importance for the calculation of dimension/ $\emptyset$ and length of steam- and condensate line.

The steam- and condensate lines are to be effected with the same $\emptyset$.

For the determination of $\emptyset$ a medium operating pressure of saturated steam of 50 bar and a steam velocity of $\leq 10\text{m/s}$ at 50 bar will be taken as basis.

The recommended tube nominal width of GEKAKONUS GMBH for the different boiler of type NUK-HP are listed in a separate table (see page 5).

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For required tubing length between the boiler of type NUK-HP and the heat exchanger for heating of vegetable oil, the contents in the system at operating pressure max. (9) minus the boiler contents (3) is decisive.

Contents in the system at operating pressure max.	=		litre	(9)
minus boiler contents	=		litre	(3)
Tube contents at operating pressure max. (from level ∇^{---2} to ∇^{---4})	=		litre	(10)

Required tubing length, steam- and condensate line are calculated from the contents each meter tubing divided by the tube contents (10).

Total tubing-length form level ∇^{---2} to level ∇^{---4}				
	Tube contents (10)	=		meter
	Contents per meter tube			
Total tubing-length form level ∇^{---2} to level ∇^{---4}		=		meter (11)

The total tubing length will be divided to each 50% to the steam – and 50% to the condensate line.

The calculated level ∇^{---4} for reason of safety function of steam generating system shall be min. 0,5 meter below the condensate drain of heat exchanger.

To avoid the likelihood of water entering into the coils of the heating tray, an additional pipe length must be installed (i.e. additional to the 87 meters calculated above). This must be installed such that there is 0.5 meters of empty vertical pipe distance between the water at operational conditions (level 4) and the outlet of the condensate drain of the coil in the heating tray level 5).

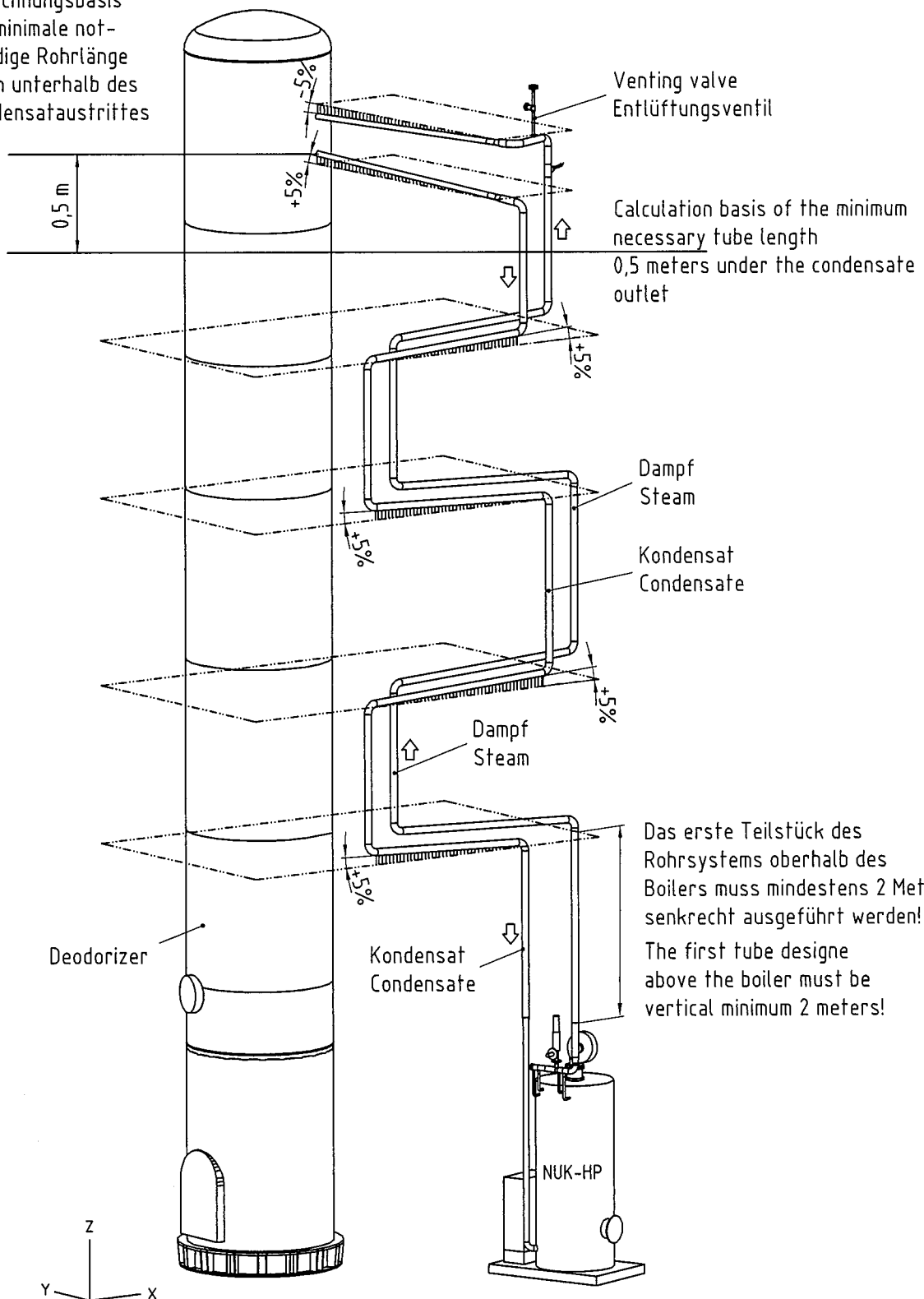
The condensate drain of heat exchanger, level ∇^{---5} shall be arranged min. 2 meter above the boiler type NUK-HP, level ∇^{---2}

If in practice, the required length of piping between level ∇^{---2} and ∇^{---4} can not be installed, or if the required minimum system volume cannot be reached, a water separator (or water buffer vessel) needs to be installed in the system (see Appendix A).

However it must be ensured, that the total tubing length (11) in the area of level ∇^{---2} and ∇^{---4} can be installed.

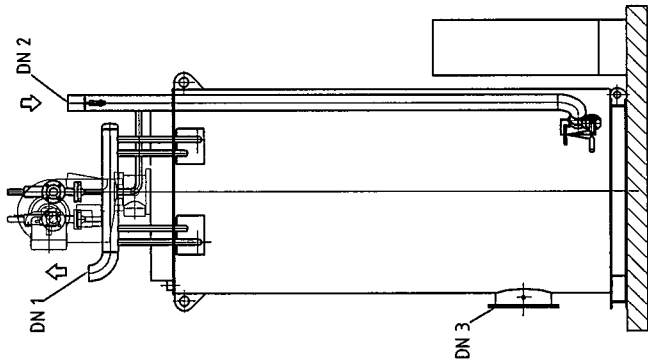
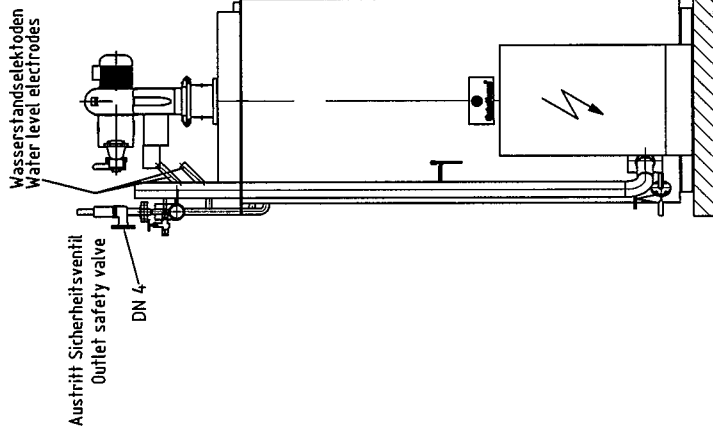
In case it should be not possible, to find a solution of problem on the strength of above mentioned described procedure, we please ask you, to contact GEKAKONUS GMBH.

Berechnungsbasis
für minimale not-
wendige Rohrlänge
0,5 m unterhalb des
Kondensataustrittes

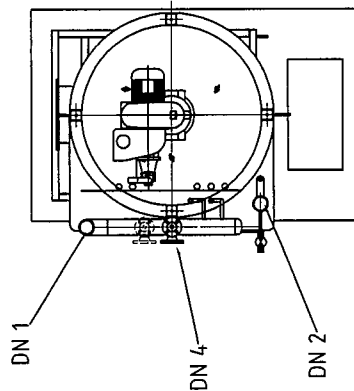
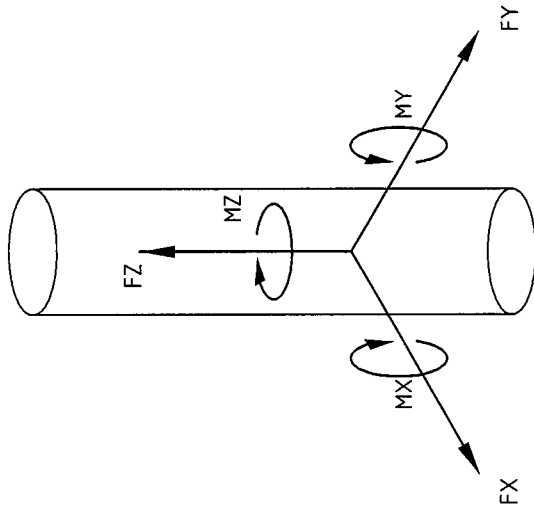


Revision		01.04.2005 / Lehmann	Klass.-Code	:	27-06	Teile Nr.	:			
						Part list No.	:			
Revision		17.12.2004 / Lehmann	Format	:	A4	Massstab:	:			
						Scale	:	%		
Erstausgabe 1st Issue		11.03.2004 / V.Gutschmidt	Genehmigt Approved	:	/	Erstellungsjahr Year of 1st issue	:	2004		
 GekaKonus® GekaKonus GmbH Siemensstr. 10 D - 76344 Eggenstein-Leop.			Titel Title		Rohrleitungsführung NUK-HP Example for pipe work NUK-HP			Disketten Nr. Disk Nr.	:	
								Dok.Nr./Doc.No.=File name		Rev.
			Schutzvermerk nach DIN 34 beachten Respect remarks acc. to DIN 34			154246				C

maximum admissible forces / moments on the pipe connection of the boiler



Pipe size DN1	FX=FY=FZ [N]	MX=MY=MZ [Nm]	Pipe size DN2	FX=FY=FZ [N]	MX=MY=MZ [Nm]
DN80 (3 inch) 80 / 95 bar	500	100	DN80 (3 inch) 80 / 95 bar	3000	1500
DN100 (4 inch) 80 / 95 bar	800	500	DN100 (4 inch) 80 / 95 bar	5000	2500
DN150 (6 inch) 80 / 95 bar	1000	700	DN150 (6 inch) 80 / 95 bar	6000	3000



- | | |
|--------|---|
| DN 1 - | Dampf – Schweißende
Stream – weld on end |
| DN 2 - | Kondensat – Schweißende
Condensate – weld on end |
| DN 3 - | Rauchrohranschluss mit Gegenflansch
Flue gas connection with counter flange |
| DN 4 - | DN50-PN4.0 ohne Gegenflansch – DIN 2635
DN50-PN4.0 without counter flange – DIN 2635 |

[illegible]

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Appendix H

Operation instructions for pump

Manufacturer:

Speck

Type:

P 11/15-150

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The GEKAKONUS GMBH NUK-HP is working in a hermetically steam-/condensate system without automatic levelling feedwater supply. The provided feed water refilling pump is only used for charging and re-charging the plant. The re-charging is manually operated and required during a low-water faultbreaking, caused by water shortage. The refilling pump can also be used for the pressure test of the system with cold water.

ATTENTION !

The charging and re- charging of the plant must only be done during pressureless operating conditions. Based on a regular installation, a simultaneous operation of feed water pump and burner is not possible. The electrical equipment for the operation of feed water filling pump are reciprocally locked in the switchboard. The electrical connection of the pump happens via a socket, installed at the outer side of the switchboard. The power supply of the socket happens via a socket at the switchboard.

The pump is mounted - ready for operation - on a base frame complete with an electrical motor, an electrical cable with plug, a cone belt and a belt guard.

The required pipe connections to the GEKAKONUS GMBH NUK-HP are in flexible design with a high-pressure hose.

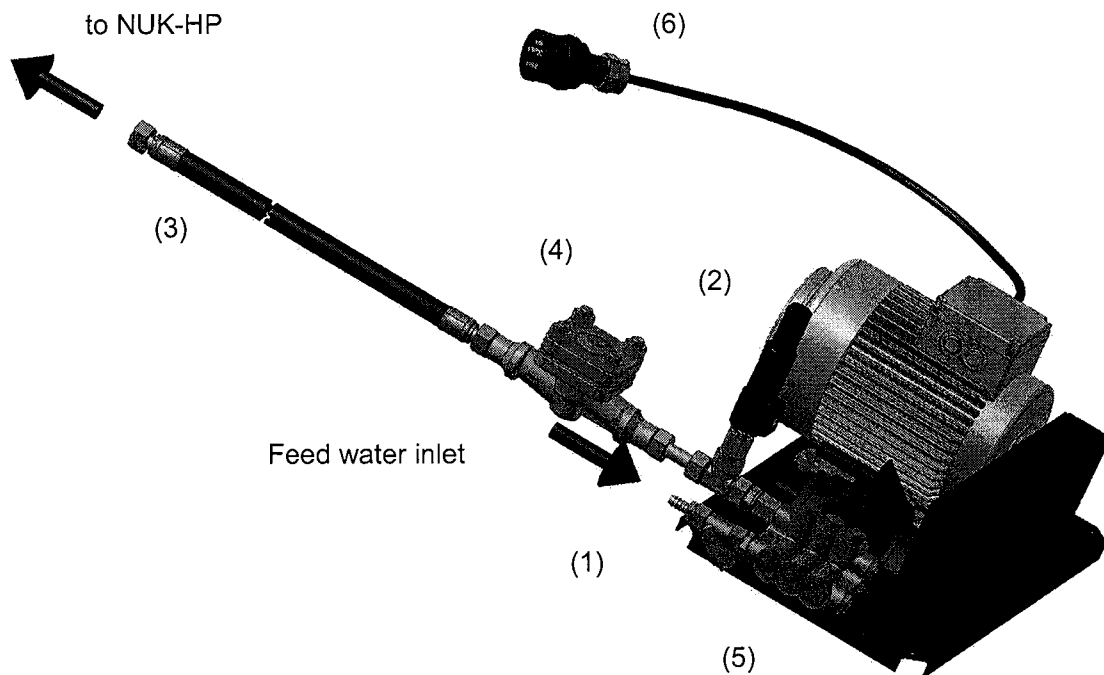
For safety reasons between high-pressure hoses and the filling/drain valve of the NUK-HP a non-return valve is installed. This prevents at burst of hoses a dangerous discharge of cold/hot water or steam.

After the charging resp. re-charging the electrical and the water side connection to the GEKAKONUS GMBH NUK-HP must be released, the filling-opening must be closed with the delivered fittings and the filling/drain valve must be closed too.

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STANDARD INSTRUMENTATION:

- 1 pc. strainer R 1/2" (1)
- 1 pc. safety valve for feed water R1/2", 140 bar (2)
- 1 pc. high pressure hose with couplings R 1/2" (3)
- 1 pc non return valve (4)
- 1 pc. plunger pump type P11/15-150 (5)
- 1 pc. electrical connection to NUK-HP switchboard (6)



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Technical Specifications

Type	P 11/15-150
Power consumption	2,2 kW
Pressure max.	150 bar
RPM max.	1420 min -1
Output max.	14,7 l/min.
Water-temp. max.	70°C
Plunger dia.	18,0 mm
Stroke	14,1 mm
Weight approx.	6,3 kg kg
NPSH Required	5,3 m.l.c.

Required NPSH refers to water : Specific weight 1 kg/dm³, viscosity 1°E at max. permissible revolutions.

Operation and Maintenance

Check oil level prior to starting and ensure trouble-free water supply. Oil : Use only 0.26 litres of ISO VG 220 (e.g. Aral Degol BG 220) or SAE 90 gear oil.

Initial change after 50 operating hours and then every 500 operating hours, after 6 months operation in any case.

ATTENTION!

When operating in damp places or with high temperature fluctuations. Oil must be changed immediately, should condensate (frothy oil) occur in the gear box. Max. Input pressure 10 bar. Keep NPSH under control.

Safety and Warranty rules

There is a 6 months warranty in accordance with VDMA guidelines.

Pump operation without safety valve as well as any excess in temperature or speed limits automatically voids the warranty. The safety valve must be regulated in accordance with the guidelines for liquid spraying units so that the admissible operating pressure can not be exceeded by more than 10 %.

When the pump is in operation, the open shaft end must be covered up by shaft protector (17), the driven shaft side and coupling by a contact protector.

Pressure in discharge line and in pump must be at zero before any maintenance to the pump takes place. Close up suction line. Disconnect fuses to ensure that the driving motor does not get switched on accidentally.

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ATTENTION!

Make sure that all parts on the pressure side of the unit are vented and refilled, with pressure at zero, before starting the pump. In order to prevent air, or an air/water-mixture being absorbed and to prevent cavitation occurring, the pump NPSHR, positive suction head and water temperature must be kept under control. Cavitation and / or compression of gases lead to uncontrollable pressure-kicks which can ruin pump and unit parts and also be dangerous to the operator or anyone standing nearby.

Plunger Pumps are suitable for pumping clean water and other non-aggressive or abrasive media with a specific weight similar to water. On no account may inflammable or explosive media be pumped.

MAINTENANCE

To check valves

Suction Valve: Screw out plugs (41). Take out suction valve adapter (39) together with suction valve. Push valve parts out of suction valve adapter using a soft tool. Check and replace worn parts.

Check O-rings (38,40,42) and replace as necessary. Tighten plugs (41, 43) to 70 NM.

To check Seals and Plunger Pipe

Screw out plugs (41). Loosen nuts (46) and remove valve casing from plungers, pulling it out to the front. Take out suction valve adapter (39), tension spring (33) and seal-unit (30,31,32). Check surfaces of plunger pipes as damaged surfaces cause fast wear to the seals.

When replacing V-sleeves (31), grease new seals with special grease from pump manufacturer before installing. Check O-rings (40,42) and replace as necessary.

If plunger pipe (24A) has to be replaced, loosen tension screw (24B) and remove it together with the plunger pipe. Check and clean plunger (22) surface and install new plunger pipe. Cover thread of tension screw with a fine film of liquid glue and tighten carefully to 22,5 NM.

ATTENTION !

Care must be taken that no glue gets between the plunger pipe (24A) and centring on plunger (22). The plunger pipe should not be strained by eccentric tightening of tension screw nor through dirt or damage to the front surface of the plunger as this could cause the plunger pipe to break.

Install tension spring (33) and suction valve adapter (39), then tighten plug (41) to 70 NM.

Fix valve case by tightening nuts (46) evenly to 47,5 NM.

To check drip return

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After removing the valve casing, the V-sleeves (50) in the intermediate casing (48) can be examined and replaced if necessary. Particular care must be taken that the two bores in the valve casing are free of lime deposits so as not to prevent the drip return.

To dismantle gear

Drain oil after dismantling the valve casing (29) and intermediate casing (48), then screw off crankcase cover (3) and bearing cover (12). Whilst turning slightly, hit out the crankshaft to one side with a rubber hammer. Do not force the cranks on the shaft when pulling through the connecting rod. If necessary, either press out second bearing in crankcase or hit it out carefully with a soft tool.

ATTENTION !

Connecting rods are marked for identification. Con rod is to be reinstalled in the same position on shaft journals. Do not bend the con rod shanks. Check shaft and con rod surfaces, shaft seals (26), crosshead + plunger (22) and ball bearings.

To reassemble

Using a soft tool, press one ball bearing in till it reaches the edge of the bearing hole. Push shaft with the other bearing in carefully through the opposite bearing hole and fit in carefully through the con rods. Press on bearings on both sides. Carefully thread shaft seal onto shaft and press into casing. Fit on bearing cover (12).

ATTENTION !

After assembly has been completed, the shaft should turn easily with very little clearance.

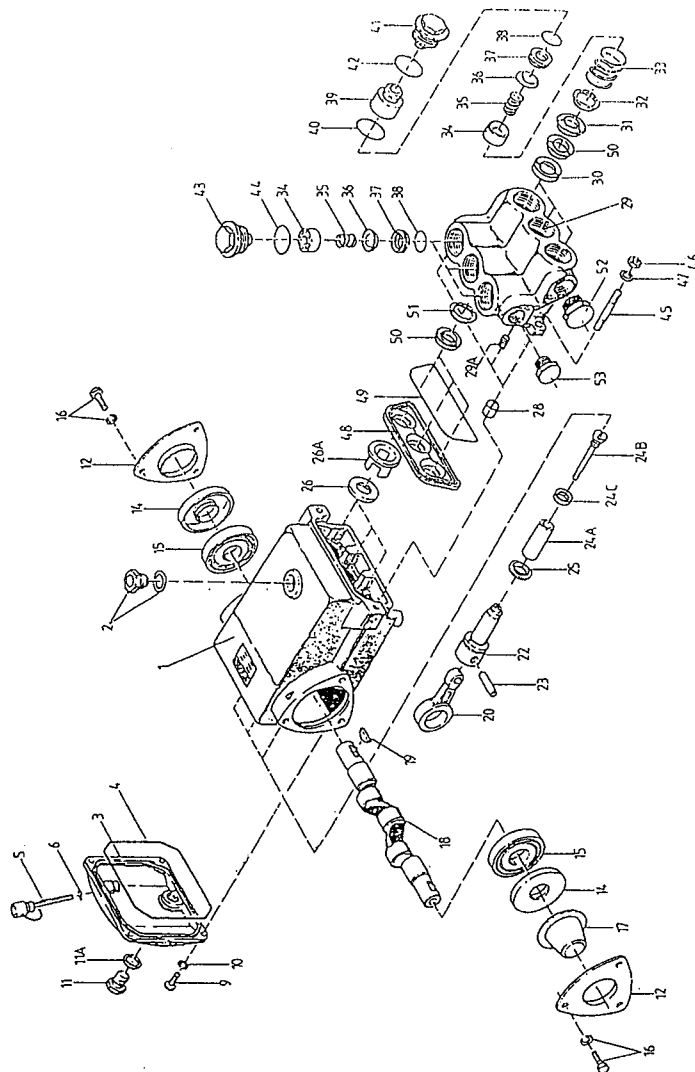
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Technische Änderungen vorbehalten
 Subject to change
P11 D

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Spare parts list P 11 / 15-150

Item no.	No. Off	Code no.	Description
1	1	01.0269	Crankcase
2	1	00.2373	Vent / Filler Plug Assy
3	1	03.0121	Crankcase Cover
4	1	06.0050	O-Ring for 3
5	1	00.0901	Oil-Dipstick Assy
6	1	06.053	O-Ring for 5
9	4	21.0069	Cylinder Screw
10	4	07.0595	Spring Ring
11	1	07.0487	Oil Drain Plug
11A	1	06.0054	Gasket
12	2	03.0159	Bearing Cover
14	2	06.0055	Radial Shaft Seal
15	2	05.0076	Grooved Ball Bearing
16	6	21.0028	Hexagon Screw
17	1	07.2530	Shaft Protector
18	1	11.0351	Crankshaft (P11/10-100)
18	1	11.0107	Crankshaft (P11/13-100)
18	1	11.0255	Crankshaft (P11/15-150)
19	1	07.0596	Woodruff Key
20	3	16.0018	Connecting Rod
22	3	00.0597	Crosshead Assy
23	3	11.0066	Crosshead Pin
24A	3	11.0115	Plunger Pipe
24B	3	21.0351	Tension Screw
24C	3	06.0306	Copper Ring
25	3	07.2513	Oil Scraper
26	3	06.0064	Radial Shaft Seal
26A	3	072964	Spacer Sleeve.
28	2	07.0558	Centring Sleeve
29	1	01.0425	Valve Casing
30	3	07.0879	Pressure Ring

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31*	3	06.1133	Sleeve
32	3	07.0880	Support Ring
33	3	07.0881	Pressure Spring
34**	6	07.3087	Spring Tension Disc
35**	6	07.1941	Valve Spring
36**	6	07.1004	Valve Plate
37**	6	07.0292	Valve Seat
38**	6	06.0067	O- Ring
39	3	07.0882	Suction Valve Adapter
40*	3	06.0271	O- Ring
41	3	07.0878	Plug M 28x1.5
42*	3	06.0272	O-Ring
43	3	00.1514	Plug G 3/4
44*	3	06.0496	O- Ring
45	2	21.0078	Stud Bolt
46	2	07.2398	Hexagon Nut
47	2	07.2706	Disc
48	1	01.0249	Intermediate Casing
49*	1	06.0869	O- Ring
50*	3	06.1057	Sleeve
51*	3	07.0941	Support Ring
52	1	07.0705	Plug G 1/2
53	1	07.0608	Plug G 3/8
	1	00.1963	Crankcase Assy P 11/15 (1-28/45-47)
	1	00.0785	Pump Head Assy (29-44/52/53)
*	1	14.0494	Seal Repair Kit
**	1	14.0414	Valve Repair Kit

When ordering please state Code no., Pump Model and Pump Serial no.

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Appendix I

Level control electrode NRG 19-50 and level switch NRS 1-50

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1.0 LEVEL CONTROL ELECTRODE NRG 19-50

1.1 FUNCTION

The level electrodes NRG 19-50 are used in conjunction with level switch NRS 1-50 as water level limiters for steam boiler plants. Water level limiters switch off the heating when the water level falls below the set minimum level (low water).

1.2 DIRECTIVES AND STANDARDS

Pressure Equipment Directive (PED) 97/23/EC (only in conjunction with NRS 1-50)

EC type approved according to EN 12952-11, EN 12953-09

These directives state, among other things, the requirements made on limiting systems and equipment for steam boiler plants.

Functional Safety according to IEC 61508 (only in conjunction with NRS 1-50)

This standard describes the functional safety of safety-related electrical / electronic / programmable electronic systems.

VdTÜV Bulletin "Wasserstand 100" (= Water Level 100)

The NRG 19-50 in conjunction with NRS 1-50 are type approved according to the VdTÜV Bulletin "Water Level 100".

1.3 TECHNICAL DATA

Type:	NRG 19-50
Maximum service pressure:	100 bar (psig)
Saturation temperature:	311°C
Nominal pressure:	PN 160
Connections:	G 3/4"
Type approval-no.:	TÜV.SWB.XX-XXX

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1.4 INSTALLATION

GESTRA level- control electrodes are high-quality measuring probes and should be handled with care. Avoid subjecting the electrode tip to shocks or bending as this might cause damage to the electrode seal.

Step 1

1. Screw electrode tip (4) into measuring electrode (1) (see Appendix I, page 4).
2. Carefully determine required measuring length of electrode
3. Mark length of electrode tip (4)
4. Unscrew electrode tip (4) from measuring electrode (1) and cut tip
5. After visual inspection screw electrode tip (4) into measuring electrode (1). Slide spring (3) along electrode tip, so that its end completely enters into the small bore (2)

Step 2

1. Check the seating surface of the thread or flange provided on the vessel or boiler standpipe
2. Place ring joint (5) onto seating surface (6) of electrode
3. Apply a light smear of silicone grease (e.g. WINIX 2150) to electrode thread (7)
4. Screw level electrode into threads provided on the boiler and tighten with a 41 mm open-end spanner. The torque required is 160 Nm when cold

Piece of advice:

- The seating surface of the thread or flange provided on the vessel of the boiler standpipe must be accurately machined
- Do not bend the electrode tip when mounting
- Do not lag the electrode body situated above the hexagonal part

1.5 INSPECTION AND TESTING

The electrode is of self-testing type, if a test failed this mail function will be indicated. For more details see next chapter (Level switch NRS 1-50)

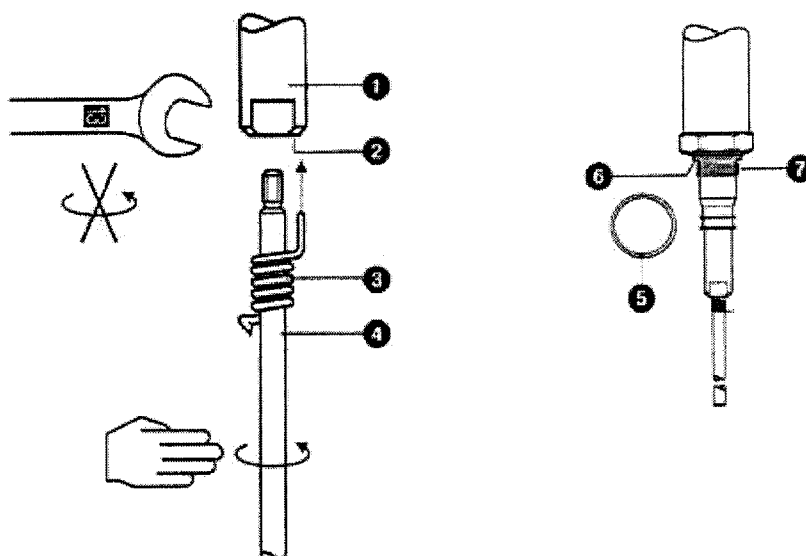
1.6 OPERATION

During the normal operation the green LED "Operation" indicates a normal function of the electrode. If the water is too low or there is a mail function of the electrode the red alarm light is on. (see description level switch NRS 1-50)

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1.7 MAINTENANCE

No user serviceable parts are contained within the equipment. Only the manufacturer must perform all repairs. Misuse or any attempted modification of the equipment will lead to considerable safety risks.



- ❶ Measuring electrode
- ❷ Bore
- ❸ Spring
- ❹ Electrode tip
- ❺ NRG 1...-50: Joint ring 27 x 32, form D, DIN 7603, 1.4301, bright annealed
NRG 111-50: Joint ring 33 x 39, form D, DIN 7603, 1.4301, bright annealed
- ❻ Seating surface
- ❼ Electrode thread

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2.0 LEVEL SWITCH NRS 1 - 50

2.1 FUNCTION

The level switch NRS 1-50 in conjunction with the NRG 19-50 is self-monitoring water level limiter with periodical self-test. This combination monitors the lowest water level in the boiler and switches off the heating when the water level falls below the set minimum level.

2.2 TECHNICAL DATA

Safety Integrity Level SIL	Classification of the Safety Integrity Level acc. to IEC 61508 SIL 3 (only in conjunction with NRG 19-50)
Safe Failure Fraction SFF	Percentage of failures without the potential to put the safety-related system into a dangerous state 97,80% (in combination with 2 electrodes NRG 19-50)
Type- approval no.:	TÜV-SWB.XX-422
Output safety circuit:	Two volt free make contacts, 24 V DC – 250 V AC, max. 6A (resistive/inductive), contact material AgNi. Delay of response: 3 sec.
Signal output:	2 volt-free outputs for instantaneous external signalling, 24-230 V AC/DC, max. 100 mA
Response sensitivity:	> 10...<10000 µS/cm at water temperature of 25°C
Indicators and adjusters:	2 buttons for test and diagnosis, 4 red/green LEDs for indicating the operation mode and alarm. 6 red LEDs for fault diagnosis, 2 two-pole code switches for setting the number of electrodes.
Mains supply:	See Wiring diagram
Protection:	Housing: IP 40 to EN 60529 Terminal strip: IP 20 to 60529
Perm. ambient temperature:	0...55°C

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2.3 INSTALLATION

On supporting rail (with mounting clip)

- Snap the level switch onto the supporting rail

Wiring should be carried out in accordance with the wiring diagram for the project (Appendix J).

Connect screens of electrode wires only to terminals 5 and 13.

At operation with two level electrodes NRG 19-50 have to be the code switches (4) and (5) in position "OFF"!

2.4 TESTING

A low-level alarm can be simulated by pushing the button "1" or "2" whilst the electrode tip is submerged. Red LED of level electrode is flashing until the response delay has expired. After this the red LED has to illuminate. The test is finished.

2.5 OPERATION

When the green LEDs illuminate during the normal operation then the water level is enough. The red LEDs illuminate when the water level falls below the minimum level or there is a fail function of the electrode.

2.6 MAINTENANCE

No user serviceable parts are contained within the equipment. Only the manufacturer must perform all repairs. Misuse or attempted modification of the equipment will endanger plant safety.

2.7 TROUBLESHOOTING

Before carrying out the fault diagnosis please check:

Mains supply:

Is the level switch supplied with the mains voltage specified on the name plate?

Wiring:

Is the wiring in accordance with the wiring diagram and the relevant schematic representation of arrangement?

Configuration:

Are the code switch settings (4) and (5) correct for the number of level electrodes used?

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Diagnosis			
Display 1 and activity	Display 2	Fault	Remedy
LED alarm 1 and diagnosis LED 1 illuminated. Press and hold down key 1 .	Diagnosis LED1 flashing.	Malfunction in level electrode 1, malfunction in level switch, faulty wiring, faulty measuring voltage.	- check wiring, - measure electrode voltage - clean and if necessary exchange level electrode - exchange level switch
	Diagnosis LED2 flashing.	Malfunction in level electrode 1, malfunction in level switch, faulty wiring.	
	Diagnosis LED3 flashing.	Interference voltage causing malfunction, boiler earth without PE.	Provide screen and earthing, connect boiler with PE.
LED alarm 2 and diagnosis LED 2 illuminated. Press and hold down key 2 .	Diagnosis LED1 flashing.	Malfunction in level electrode 1, malfunction in level switch, faulty wiring, faulty measuring voltage.	- check wiring, - measure electrode voltage - clean and if necessary exchange level electrode - exchange level switch
	Diagnosis LED2 flashing.	Malfunction in level electrode 1, malfunction in level switch, faulty wiring.	
	Diagnosis LED3 flashing.	Interference voltage causing malfunction, boiler earth without PE.	Provide screen and earthing, connect boiler with PE.
LED alarm 1 and 2 and diagnosis LED 3 illuminated. Press and hold down key 1 or 2 .	Diagnosis LED1 flashing.	Malfunction in processor, stand-by fault.	Observe operating instructions for the logic unit SRL. Replace level switch.
	Diagnosis LED2 flashing.	Internal voltage fault.	
	Diagnosis LED3 flashing.	Malfunction in relay.	

Once the fault is eliminated, the level switch returns to normal operation.

After elimination of the fault switch off the mains voltage and switch it on again after approx. 5 sec.

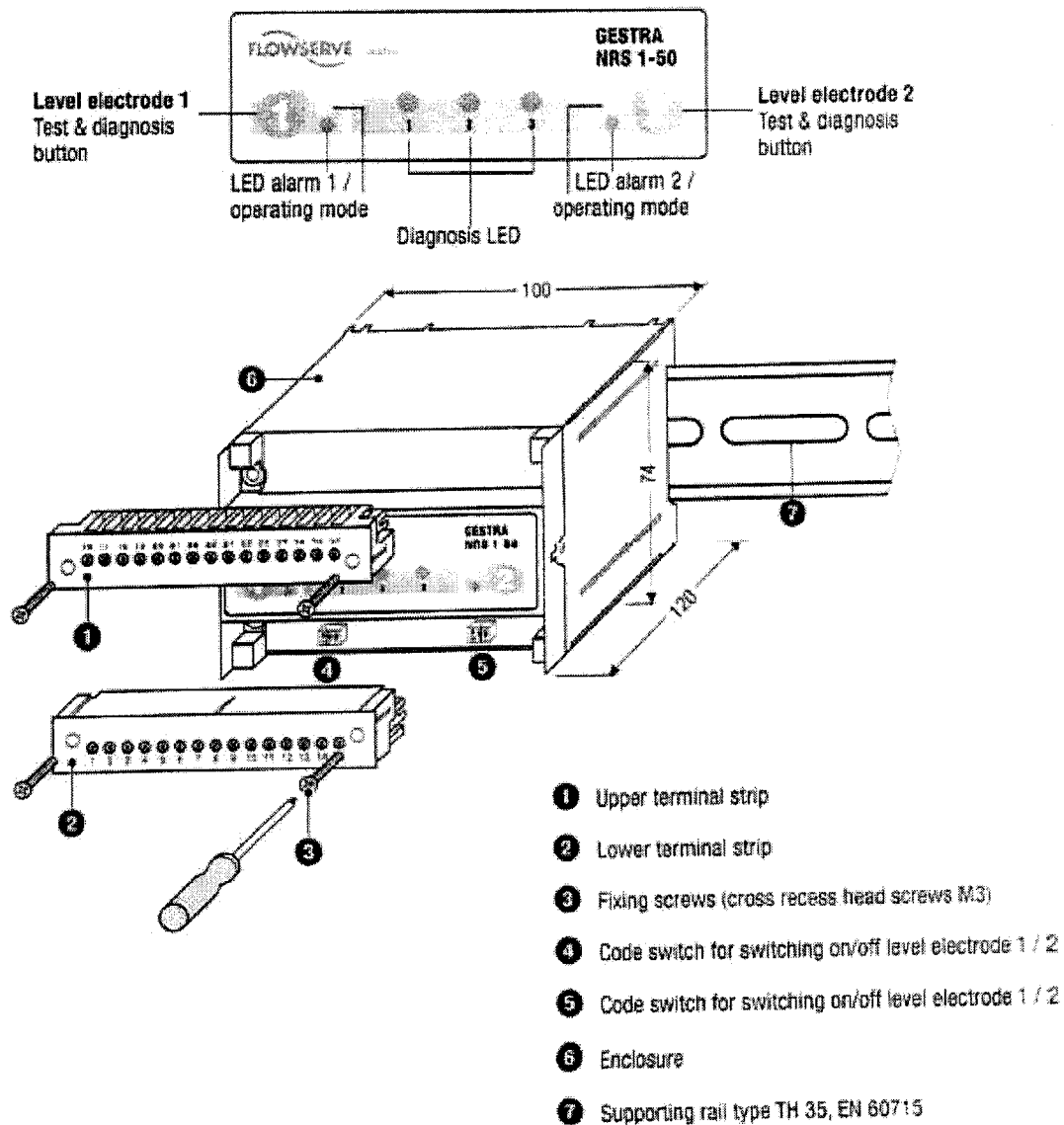
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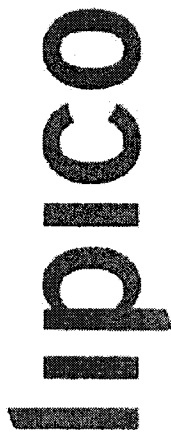
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WIRING DIAGRAM			
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Appendix J

Wiring diagram

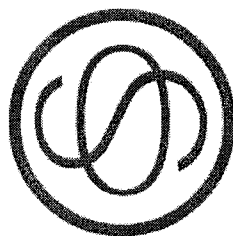
Wiring diagram

List of parts



www.pico.com

wiring diagram



Gekakonus®

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Phone +49 (0) 721 / 94374-0

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100-443887-100

customer

part of plant : NUK-HP-1500

drawing no. : 10-9852

cabinet : 1000x1400x400mm WHD

power supply	: 3x380V/50Hz
control voltage	: 220V
signal voltage	: 220V / 24V
standard	: CENELEC HD 384 / EN 50156-1 (VDE 0100) (VDE 0116)
system of protection	: IP54

designed with ELCAD-AUCOTEC (K) 7/110
SIS-Schallanlagen GmbH
Ettosengstrasse 42
D-76275 Ettlingen
D-76344 Eggenstein
www.ses-schallanlagen.de
info@ses-schallanlagen.de

[illegible]

1	2	3	4	5	6	7	8
A	B	C	D	E	F		
Erläuterungen zur Funktion Commentary for the function		Klemmen ausserhalb des Schaltschranks terminals outside of the switchboard -XB o ==> Brennerklemmen, burner terminals Der Schaltplan besteht aus Einzelblättern welche durch Blatt-Nr. und Blatt-Anzahl gekennzeichnet sind The wiring diagram is composed of separate sheets, marked with sheet-number and total number of the sheets.					
Jedes Blatt ist in einzelne Abschnitte unterteilt Each sheet is subdivided in sections		Alle Geräte erhalten als Bezeichnung die Nummer des jeweiligen Abschnittes The symbol for all units is the number of the section belonging to it.					
Verbindungen (Abbruchstellen) zu anderen Blättern oder Abschnitten des Schaltplanes werden mit den Nummern der Abschnitte und des Blattes gekennzeichnet ! Connections with continuation on other sheets or sections are marked with numbers of the resp. sections !		WARNING Installation, operation and maintenance procedures performed by unqualified personnel may result in improper adjustment and unsafe operation. Either condition may result in equipment damage or personnel injury. Only qualified and specific trained personnel is allowed to carry out the installation, commissioning, operation and maintain the series boiler/heater and the boiler/heater equipment. The boiler/heater and boiler/heater equipment is tested of logic function prior the boiler/heater leave the Gekakonus manufacturing shop. All actions on the boiler/heater and boiler/heater equipment done by the own and/or foreign staff is on the risk and responsibility of the buyer and/or operator of the boiler/heater. WELD CAUTION Switch off the main switch on the switchboard and interrupt the protection earth (PE) cable before begin of any welding on the boiler/heater piping system. Do not connect the ground connection of the welding set on any part of the boiler/heater during the welding on the boiler/heater piping system. The electrical boiler/heater system is endangered and can be destroyed. Connect the protection earth (PE) cable again when the welding procedure is concluded. Gekakonus D - 75344 Engsthalen-Löpp.					
designed with ELCAD/AUCOPLAN (R) 7.11.0		Wiring diagram SES-18-3899 WUCHP 95bar cover 2 Gekakonus origin IPICO substitute for substitute through date 05.11.2018 proc. E.H. rev. name norm alteration 1 2 3 4 5 6 7 8 sheet 0A 54 Bl.					

1	2	3	4	5	6	7	8
boiler	type	sheet	name of sheet	date	special memo		
NUK-HP	1500	0	cover 1	22.05.2018			
NUK-HP	1500	0A	cover 2	22.05.2018			
NUK-HP	1500	0B	table of contents	22.05.2018			
NUK-HP	1500	0C	table of contents	22.05.2018			
NUK-HP	1500	A1	part list	22.05.2018			
NUK-HP	1500	A2	part list	22.05.2018			
NUK-HP	1500	A3	part list	22.05.2018			
NUK-HP	1500	B	arrangement drawing mounting plate	22.05.2018			
NUK-HP	1500	C	arrangement drawing panel door	22.05.2018			
NUK-HP	1500	1	feeding	22.05.2018			
NUK-HP	1500	2	emergency stop	22.05.2018			
NUK-HP	1500	3	primary circuit	22.05.2018			
NUK-HP	1500	4	primary circuit	22.05.2018			
NUK-HP	1500	5	pump control	22.05.2018			
NUK-HP	1500	6	burner motor control	22.05.2018			
NUK-HP	1500	7	cooling air fan,preheater control	22.05.2018			
NUK-HP	1500	8	SPS-LOGO control	22.05.2018			
NUK-HP	1500	9	safety circuit	22.05.2018			
NUK-HP	1500	10	safety circuit	22.05.2018			
NUK-HP	1500	11	safety circuit	22.05.2018			
NUK-HP	1500	12	safety circuit	22.05.2018			
NUK-HP	1500	13	safety circuit	22.05.2018			
NUK-HP	1500	14	safety circuit	22.05.2018			
NUK-HP	1500	15	safety circuit	22.05.2018			
NUK-HP	1500	16	digital inputs LOGO-SPS	22.05.2018			
NUK-HP	1500	17	digital outputs LOGO-SPS	22.05.2018			
NUK-HP	1500	18	switch board fan	22.05.2018			
NUK-HP	1500	19	burner control	22.05.2018			
NUK-HP	1500	20	burner control	22.05.2018			
NUK-HP	1500	21	burner control	22.05.2018			
NUK-HP	1500	22	burner control	22.05.2018			
NUK-HP	1500	23	burner control	22.05.2018			
NUK-HP	1500	24	burner control	22.05.2018			
NUK-HP	1500	25	burner control	22.05.2018			
NUK-HP	1500	26	burner control	22.05.2018			
NUK-HP	1500	27	burner control	22.05.2018			
NUK-HP	1500	28	burner control	22.05.2018			
NUK-HP	1500	29	burner control	22.05.2018			
NUK-HP	1500	30	burner control	22.05.2018			
NUK-HP	1500	31	burner control	22.05.2018			
NUK-HP	1500	32	burner control	22.05.2018			
NUK-HP	1500	33	burner control	22.05.2018			
NUK-HP	1500	34	burner control	22.05.2018			
NUK-HP	1500	35	burner control	22.05.2018			
NUK-HP	1500	36	burner control	22.05.2018			
NUK-HP	1500	37	messages	22.05.2018			
NUK-HP	1500	38	pot. free cont.	22.05.2018			
NUK-HP	1500	39	control voltage 24VDC	22.05.2018			
NUK-HP	1500	40	spare	22.05.2018			

substitute for

substitute through

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NUK-HP

1500

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	1	2	3	4	5	6	7	8
	boiler	type	sheet	name of sheet	date	special memo		
A	NUK-HP	1500	41	connection diagram -XP-X1	22.05.2018			
	NUK-HP	1500	42	connection diagram -X2-X3-X4-X6-XL-XN	22.05.2018			
	NUK-HP	1500	43	connection diagram -XS1->XS2->XS3	22.05.2018			
	NUK-HP	1500	44	connection diagram -X94-XS5-XS6	22.05.2018			
	NUK-HP	1500	45	connection diagram -XS7-XS8-XS9-XS10	22.05.2018			
B								
C								
D								
E								
F								

wiring diagram SES-18-3899
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1500
95bar
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ipico
INDUSTRIAL PIPING CONNECTION

Gekakonus*
ORIGIN

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rev.

date proc. rev.

sheet OC
54 Bl.

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6 7 8

part list

equipment identifier	item no.	class	manufacturer	Typ	comment	refer to
-A82	299310	A	SI	6ED1052-1FB00-0BA8	SIEMENS-LOGO	/8.B2
-A84	299313	A	SI	6ED1055-1FB00-0BA1	SIEMENS-LOGO Erw.8DI/8DA	/8.B4
-A85	299312	A	SI	6ED1055-1MA00-0BA0	SIEMENS-LOGO Erw.	/8.B5
-A87	299314	A	SI	6ED1055-4MH00-0BA0	LOGO-08-Text Display	/8.B7
-A191	0330000500	A	WE	W-FM 100	combustion manager	/19.A1
-A213	0330000502	A	WE	ABE	comb. manager indication-/operation unit	/21.D3
-F13	0040003200	F	FI	7P.23.8.275.1020	surge protection device	/1.D3
-F14	0040000451	F	KM	FAZ-S2/1	capacity breaker control voltage	/1.E3
-F14.1	0040000451	F	KM	FAZ-S2/1	capacity breaker control voltage	/1.E4
-F15	0040000454	F	KM	FAZ-S6/1	capacity breaker control voltage	/1.E4
-F15.1	0040000453	F	KM	FAZ-S4/1	capacity breaker control voltage	/1.E5
-F16	0040000454	F	KM	FAZ-S6/1	capacity breaker control voltage	/1.C6
-F17	0040000451	F	KM	FAZ-S2/1	capacity breaker control voltage	/1.C7
-F17.1	0040000451	F	KM	FAZ-S2/1	capacity breaker control voltage	/1.C7
-F100	123207	F	GE	NRS 1-50	water level limiter	/11.B2
-F182	400004	F	RJT	SK 3110.00	temperature controller switch board fan	/18.A2
-G395	2990100	G	MW	HDR-30-24	power supply	/39.A4
-H27	0040001330	H	KM	M22-L-G	signal lamp	/2.E7
-H55	0040001290	H	KM	M22-L-B	signal lamp	/5.D5
-H56	0040001310	H	KM	M22-L-R	signal lamp	/6.E6
-H57	0040001290	H	KM	M22-L-B	signal lamp	/5.D8
-H74	0040001310	H	KM	M22-L-R	signal lamp	/7.D4
-H76	0040001330	H	KM	M22-L-G	signal lamp	/7.D6
-H177	0230000102	H	WER	118 068 28,230V	hooter	/17.D7
-H263	0040001330	H	KM	M22-L-G	signal lamp	/26.E2
-H264	0040001330	H	KM	M22-L-G	signal lamp	/26.E4
-H275	0040001330	H	KM	M22-L-G	signal lamp	/27.D5
-H287	0040001310	H	KM	M22-L-R	signal lamp	/28.E7
-H288	0040001310	H	KM	M22-L-R	signal lamp	/28.E8
-H377	0040001330	H	KM	M22-L-G	signal lamp	/37.D7
-H378	0040001320	H	KM	M22-L-Y	signal lamp	/37.D8
-K23	0150000004	K	DO	LG 5925.02/61	emergency-stop relay	/2.C3
-K24	0041000016	K	KM	DILA-22,24VDC	auxiliary relay	/2.E5
-K25	0030000020	K	ABB	NF31E-11,24VDC	auxiliary relay	/2.E5

comments :

alteration	date	named norm	rev.	date	05.11.2018	proc.	E.H.	substitute through	substitute for	origin	Gekakonus®	wiring diagram SES-18-3899	sheet A1
												1500	54
												95bar	Bl.
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part list

equipment identifier	item no.	class	manufacturer	Typ	comment	refer to
-K26	0040003054	K	FI	55.32 9 024 00 40	auxiliary relay	/2.E6
-K53	0041000030	K	KM	DILM9-10	main contactor	/5.E3
-K55	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/5.E5
-K57	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/5.E7
-K62	0041000032	K	KM	DILM12-10	main contactor	/6.E2
-K63	0040000047	K	KM	ETR4-51-A	timing relay	/6.E3
-K64	0041000032	K	KM	DILM12-10	main contactor	/6.E4
-K65	0041000032	K	KM	DILM12-10	main contactor	/6.E5
-K72	0041000030	K	KM	DILM9-10	main contactor	/7.E2
-K73	0040003056	K	FI	55.34 8 230 50 40	auxiliary relay	/7.E3
-K75	0041000034	K	KM	DILM17-10	main contactor	/7.E5
-K77	0041000034	K	KM	DILM17-10	main contactor	/7.E7
-K97	0041000003	K	KM	DILA-22	auxiliary relay	/9.E7
-K98	0030000022	K	ABB	NF22E-13	auxiliary relay	/9.E8
-K103	0041000003	K	KM	DILA-22	auxiliary relay	/10.D3
-K104	0030000022	K	ABB	NF22E-13	auxiliary relay	/10.D4
-K127	0041000003	K	KM	DILA-22	auxiliary relay	/12.E7
-K128	0030000022	K	ABB	NF22E-13	auxiliary relay	/12.E8
-K135	0040003054	K	FI	55.32 9 024 00 40	auxiliary relay	/13.E5
-K147	0041000002	K	KM	DILA-31	auxiliary relay	/14.E7
-K148	0030000022	K	ABB	NF22E-13	auxiliary relay	/14.E8
-K152	0030000002	K	ABB	N 31E	auxiliary relay	/15.D2
-K154	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/15.D4
-K176	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/17.D6
-K197	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/19.E7
-K262	0040003056	K	FI	55.34 8 230 50 40	auxiliary relay	/26.E2
-K263	0040003056	K	FI	55.34 8 230 50 40	auxiliary relay	/26.E3
-K271	0030000022	K	ABB	NF22E-13	auxiliary relay	/27.D1
-K272	0041000056	K	KM	ETR2-12	timing relay	/27.D2
-K273	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/27.D2
-K274	00500000300	K	SI	3RT2023-1AP	main contactor	/27.D3
-K274	0050001004	K	SI	3RT2926-2PA01	timing unit cut in delayed	/27.D3
-K275	0041000030	K	KM	DILM9-10	main contactor	/27.D4
-K286	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/28.E5

comments :

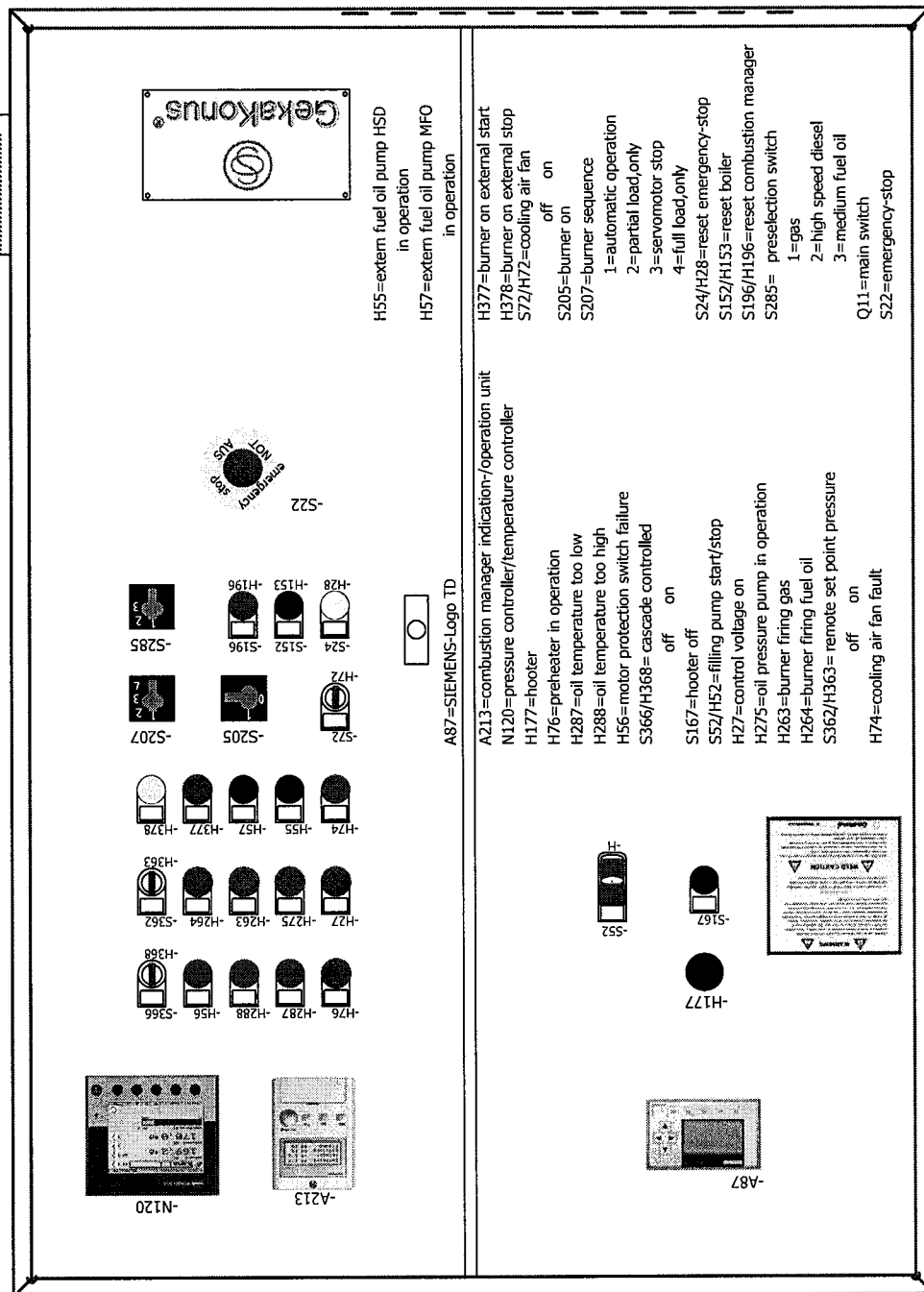
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part list

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-K323	0030000021	K	ABB	NF31E-13	auxiliary relay	/32.D3
-K324	0030000021	K	ABB	NF31E-13	auxiliary relay	/32.D4
-K334	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/33.E4
-K366	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/36.D6
-K372	0030000022	K	ABB	NF22E-13	auxiliary relay	/37.D2
-K374	0040003051	K	FI	55.32 8 230 00 40	auxiliary relay	/37.D4
-M182	400070	M	RIT	SK 3323.107/207	switch board fan	/18.D2
-N120	2304110006	N	JU	DICON IMAGO	pressure / temperature controller	/33.A2
-N278	230903	N	JU	JUMO DR100	temperature controller	/27.B5
-N286	230903	N	JU	JUMO DR100	temperature controller	/28.B3
-N304	230903	N	JU	JUMO DR100	temperature controller	/30.B1
-Q11	0070000008	Q	SO	NLT100/3E/Z33	main switch	/1.B1
-Q15	0040000094	Q	KM	PKZM0-6,3 T	motor protection switch	/1.A5
-Q32	0040000095	Q	KM	PKZM0-6,3	motor protection switch	/3.A2
-Q35	0040000093	Q	KM	PKZM0-4	motor protection switch	/3.A5
-Q37	0040000096	Q	KM	PKZM0-1,6	motor protection switch	/3.A7
-Q42	0040000098	Q	KM	PKZM 0-16	motor protection switch	/4.A2
-Q45	0040000100	Q	KM	PKZM 0-25	motor protection switch	/4.A5
-S22	0040001110	S	KM	M22-PV	emergency stop	/2.B2
-S24	0040001171	S	KM	M22-DL-Y	push button	/2.A4
-S52	0040001150	S	KM	M22-DDL-GR-GB1/GB0	double push button	/5.B2
-S72	0040001614	S	KM	M22-WRLK-G	control switch	/7.B2
-S152	0040001180	S	KM	M22-DL-B	push button	/15.A2
-S167	0040001100	S	KM	M22-D-S	push button	/16.B8
-S196	0040001170	S	KM	M22-DL-R	push button	/19.C6
-S205	0070000050	S	SO	A1/8ZM/70S	control switch	/20.C5
-S207	0070000058	S	SO	ST41/8ZM/70S	control switch	/20.C7
-S285	0070000064	S	SO	ST31/8ZM/70S	control switch	/32.B3
-S366	0040001614	S	KM	M22-WRLK-G	control switch	/36.A6
-T2	0330000341	T	WE	Trafo W-FM	transformer	/22.C3
-T15	0130000027	T	ELTRA	IST473/29,1000VA	transformer	/1.B5

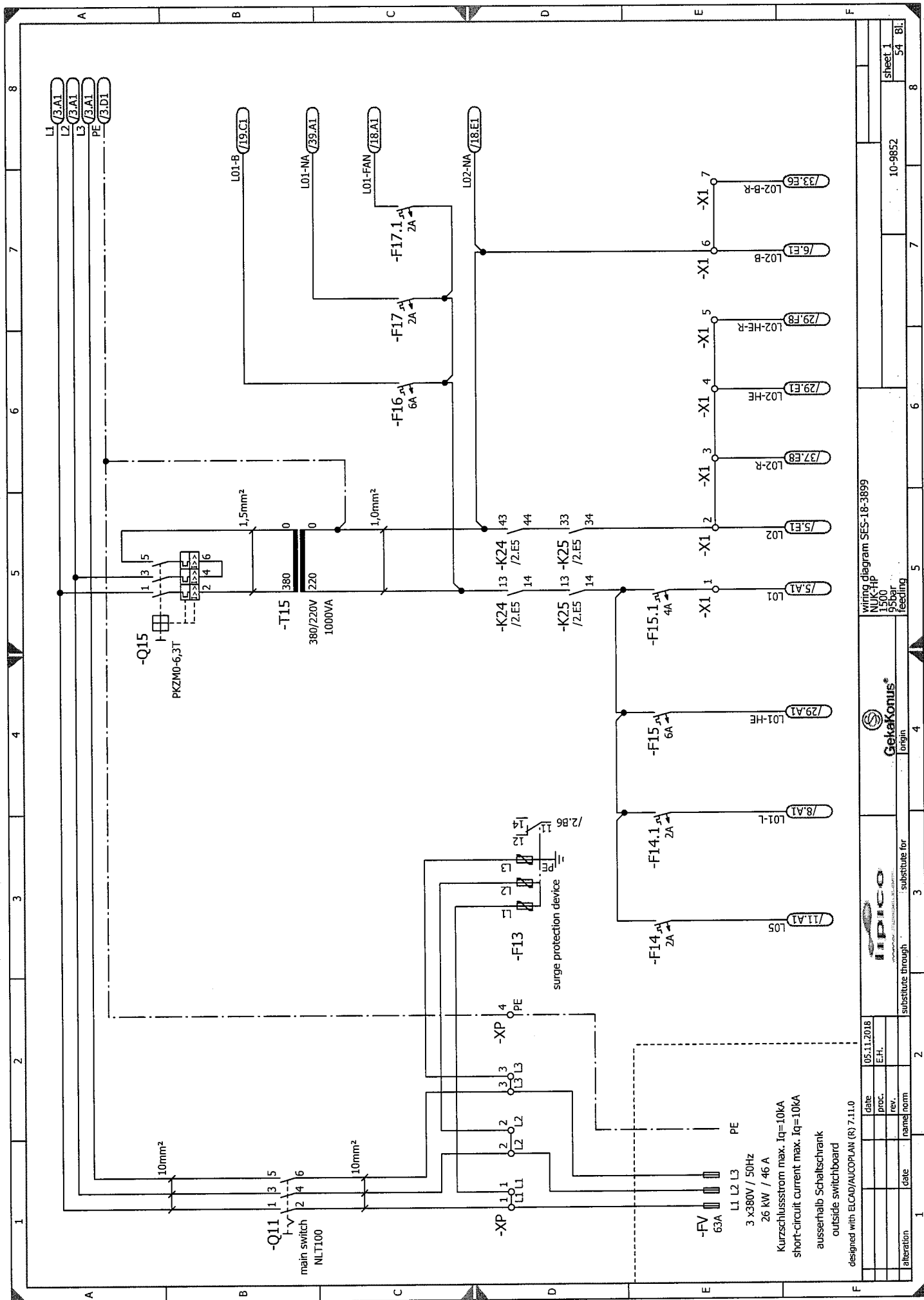
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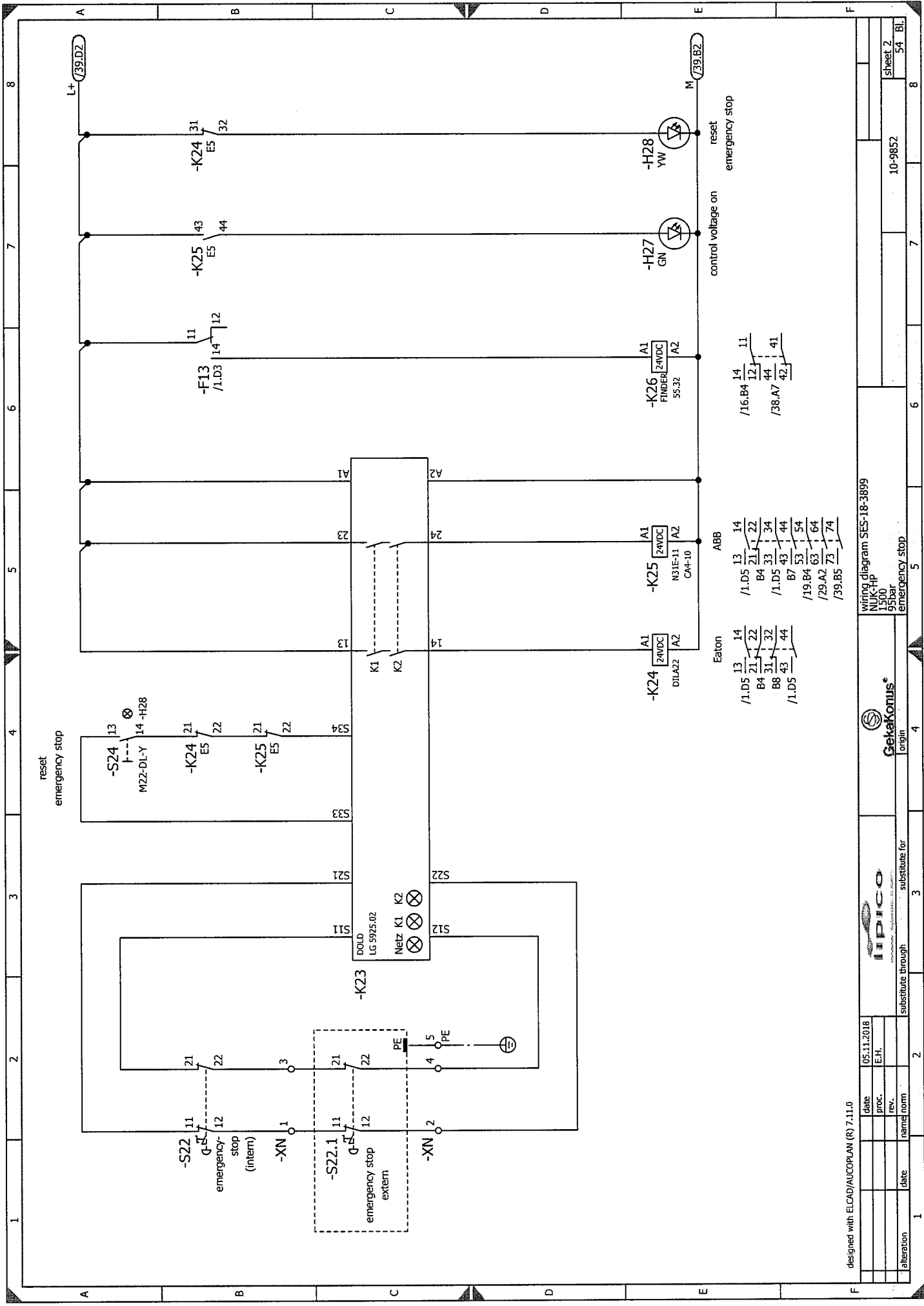
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	date	name	rev.	E.H.	substitute for substitute through origin				



scale : 1:5
measure unit : mm

[illegible]





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wiring diagram SES-18-3899



date 05.11.2018

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name norm

date

alteration

substitute through

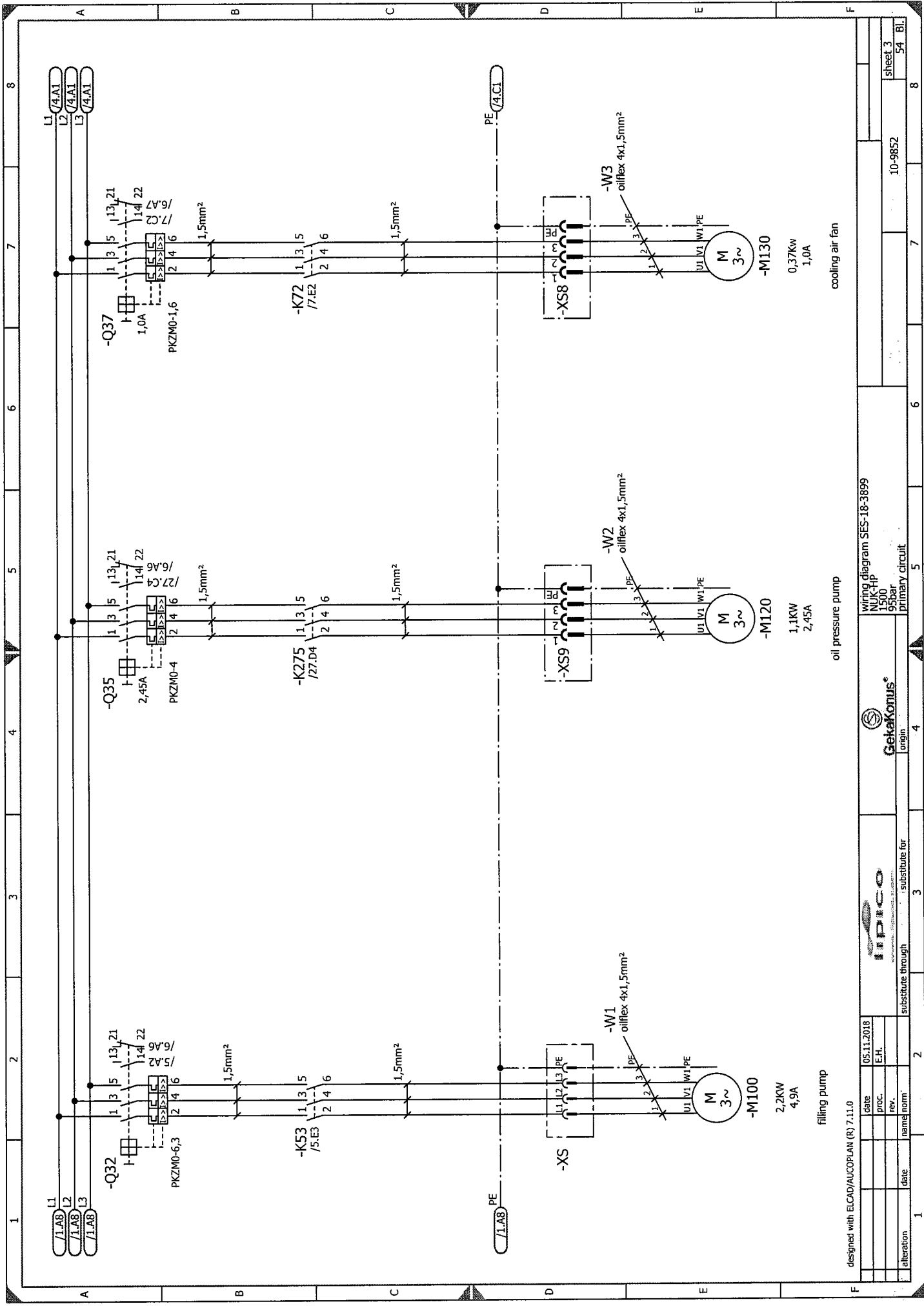
origin

emergency stop

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sheet 2

54 Bl.



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wiring diagram SES-18-3899

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INDICO

GekafKonius®

oil pressure pump

cooling air fan

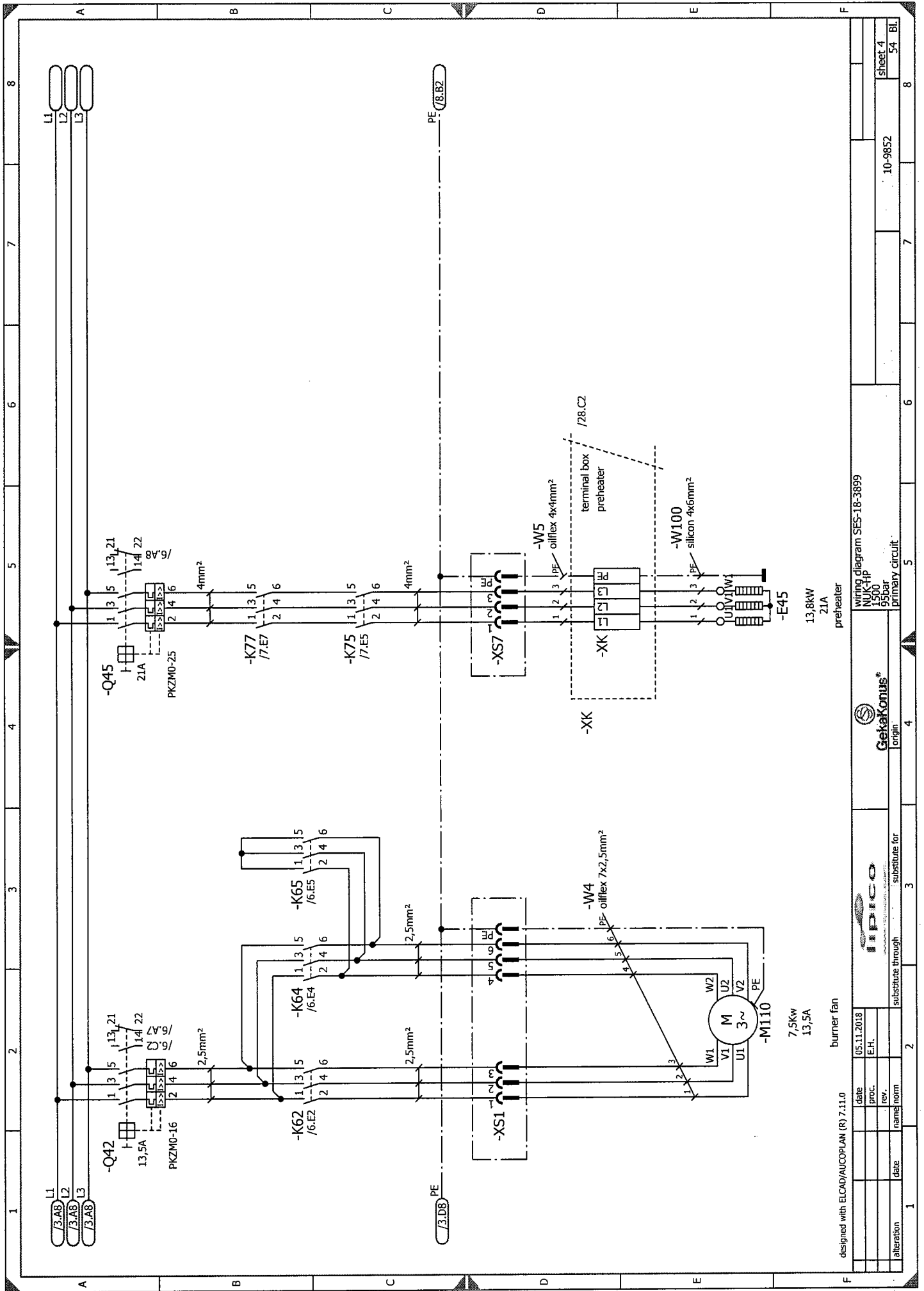
filling pump

primary circuit

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sheet 3

54 Bl.



Wiring diagram SES-18-3899

Gekakonus®
origin

substitute for

substitute through

designed with ELCAD/AUCOPLAN (R) 7.11.0

burner fan
7.5Kw
13.5A

alteration

date

named norm

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proc.

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05.11.2018

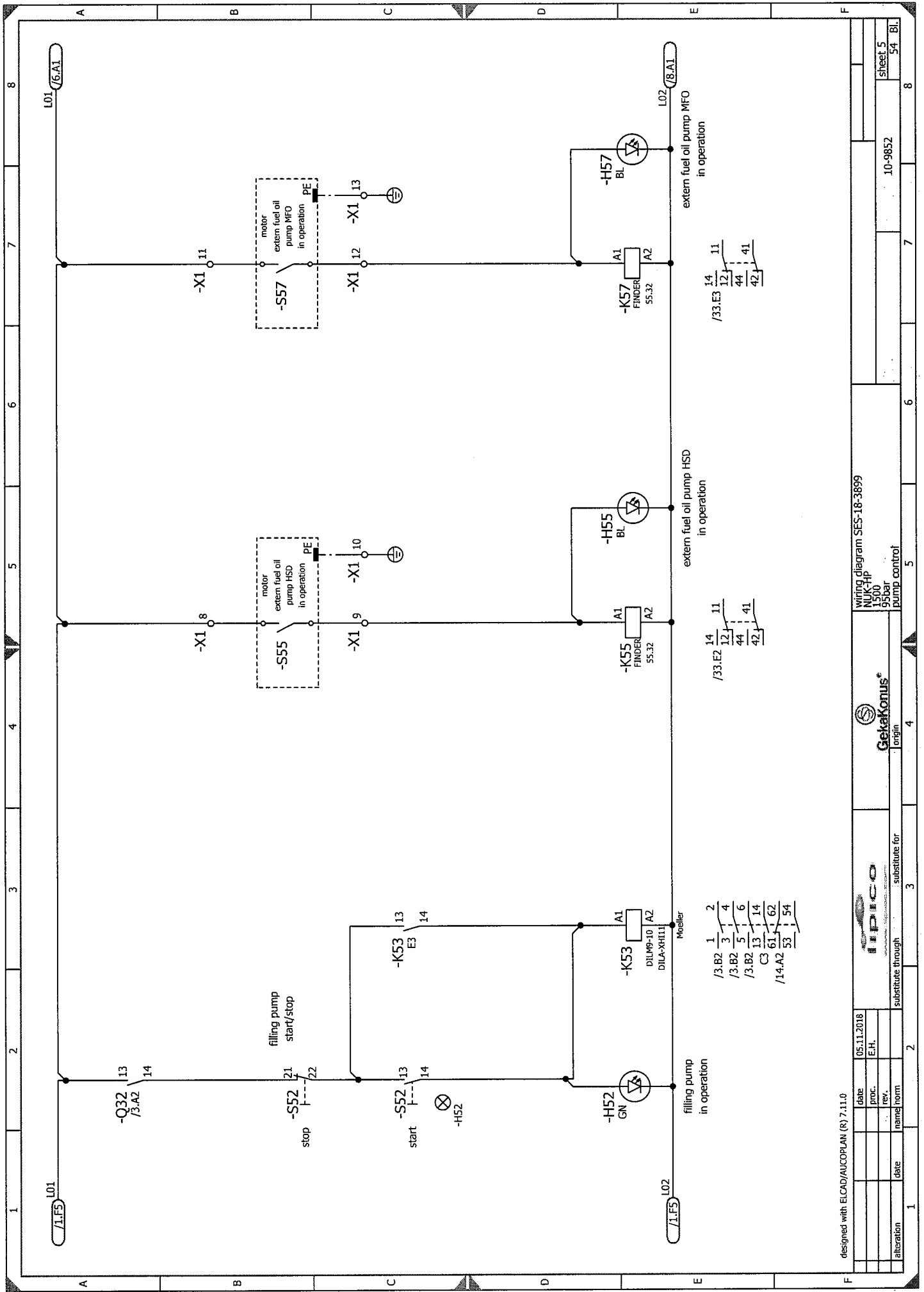
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sheet 4

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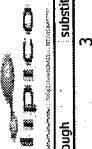
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primary circuit



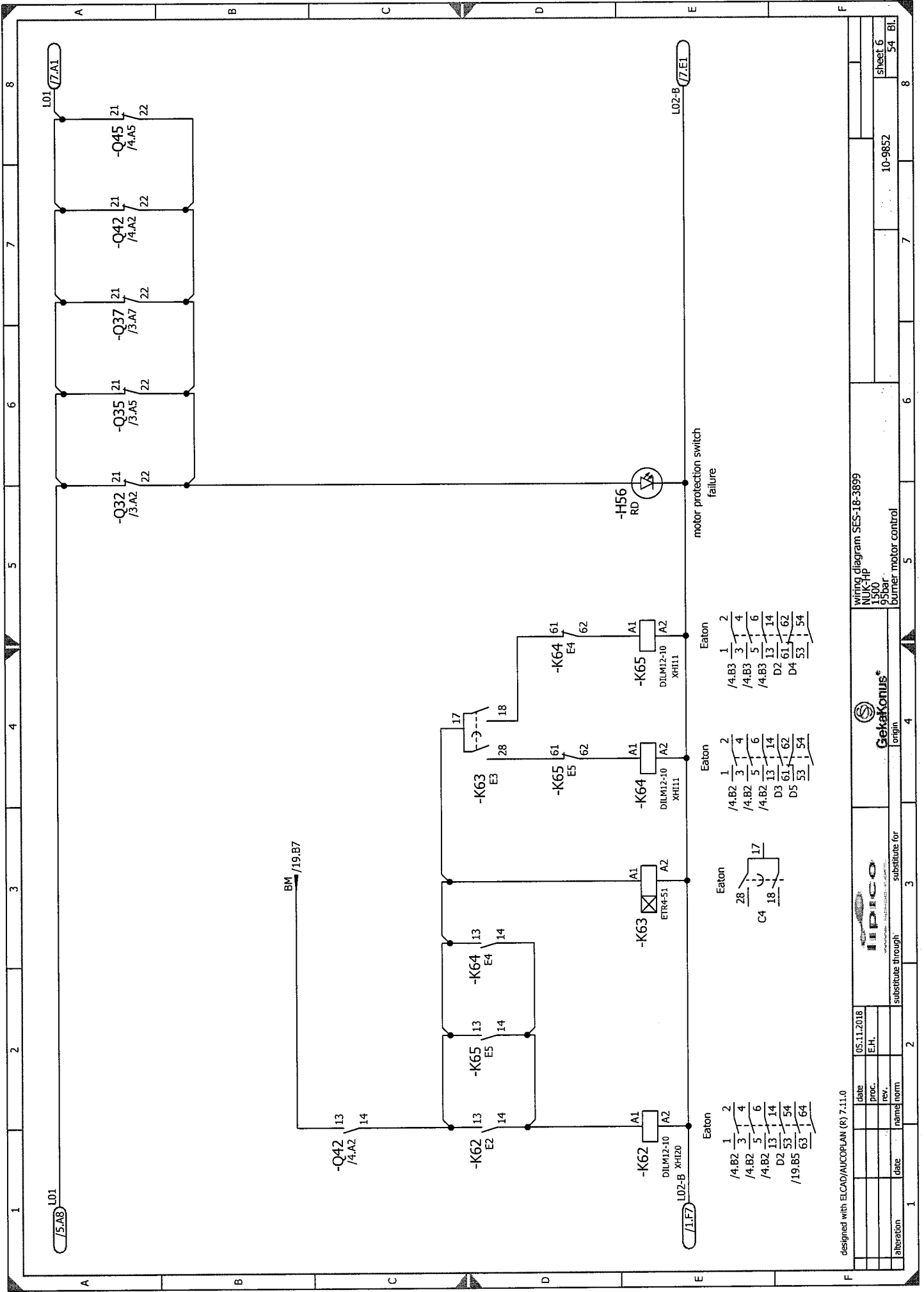
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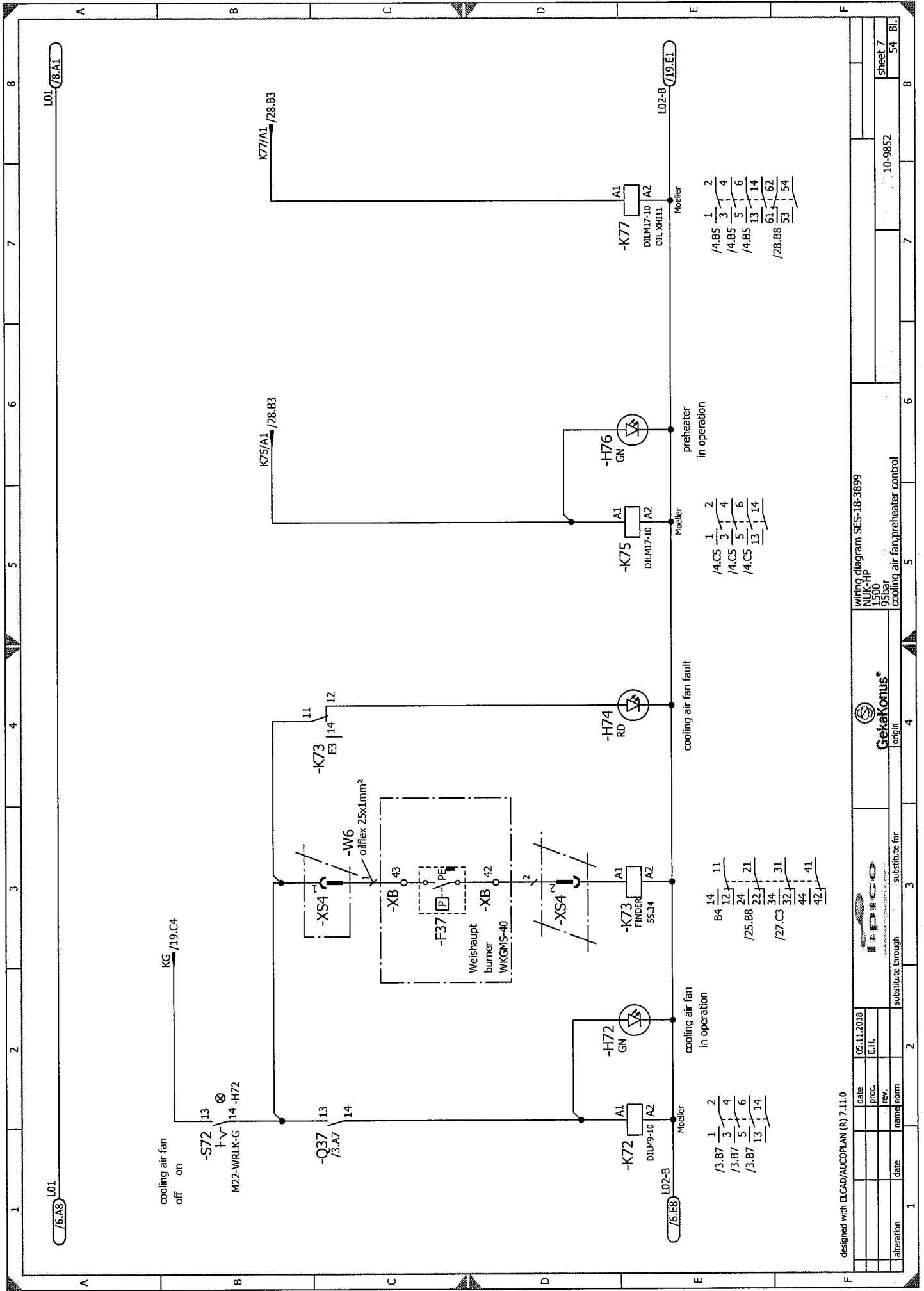
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Wiring diagram SES-18-3899
NKK-HP
1500
9500
pump control

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sheet 5
54 Bl.



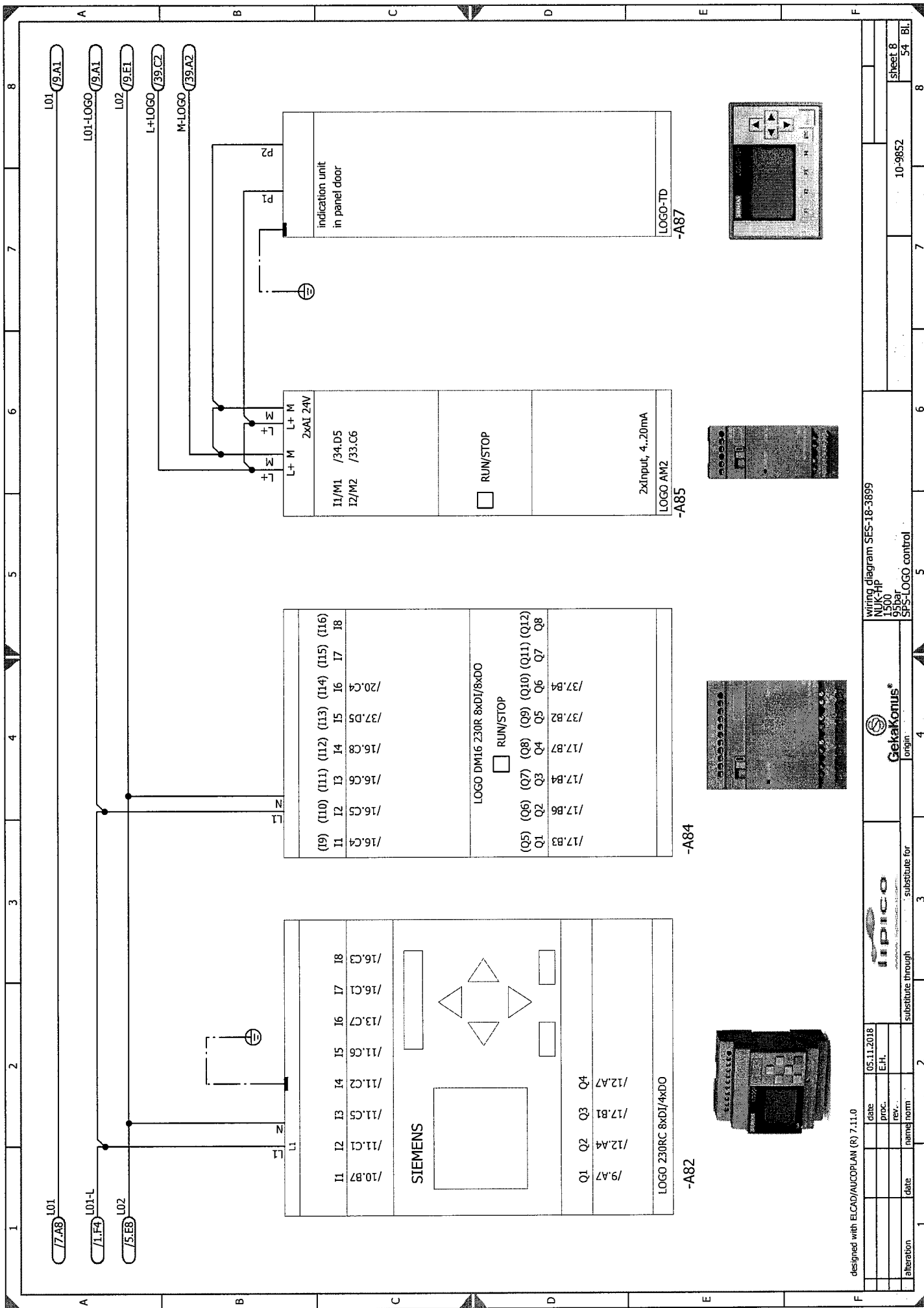


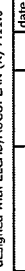
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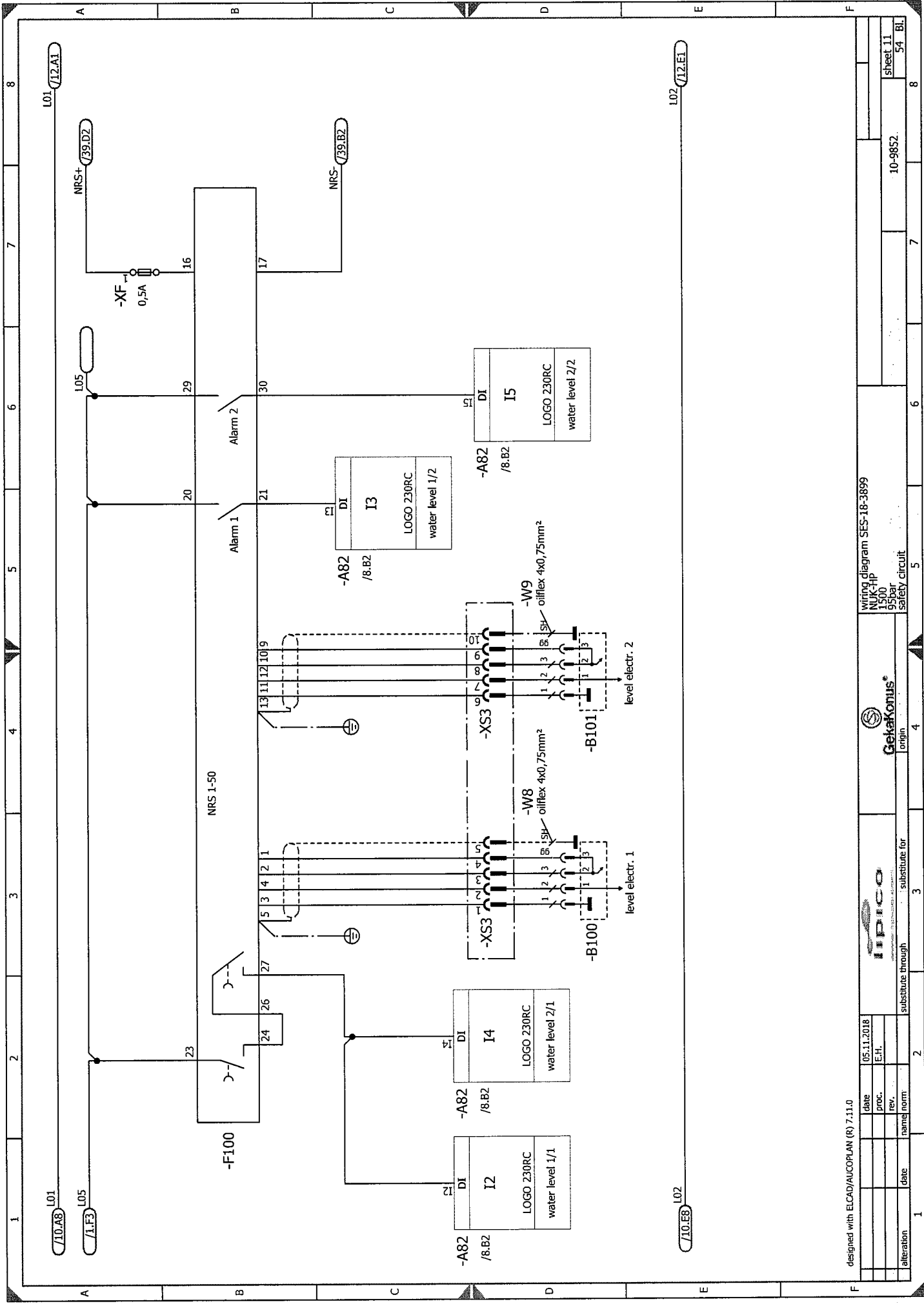


wiring diagram SES-18-3899
NKK-Hp
9500
cooling air fan, preheater control

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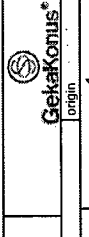




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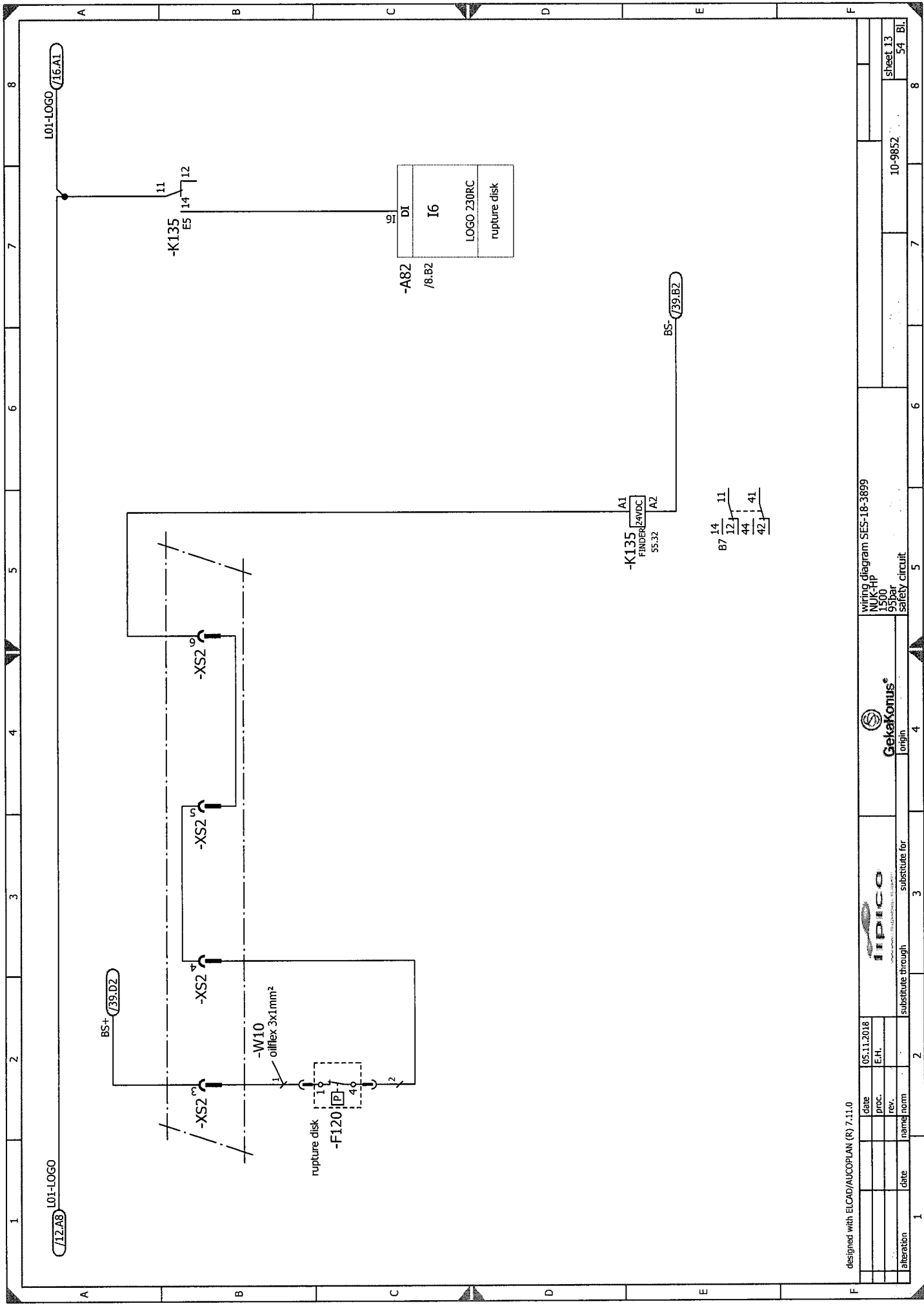
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wiring diagram SES-18-3899

100-11p
100-11p
950p
safety circuit

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54 BL	



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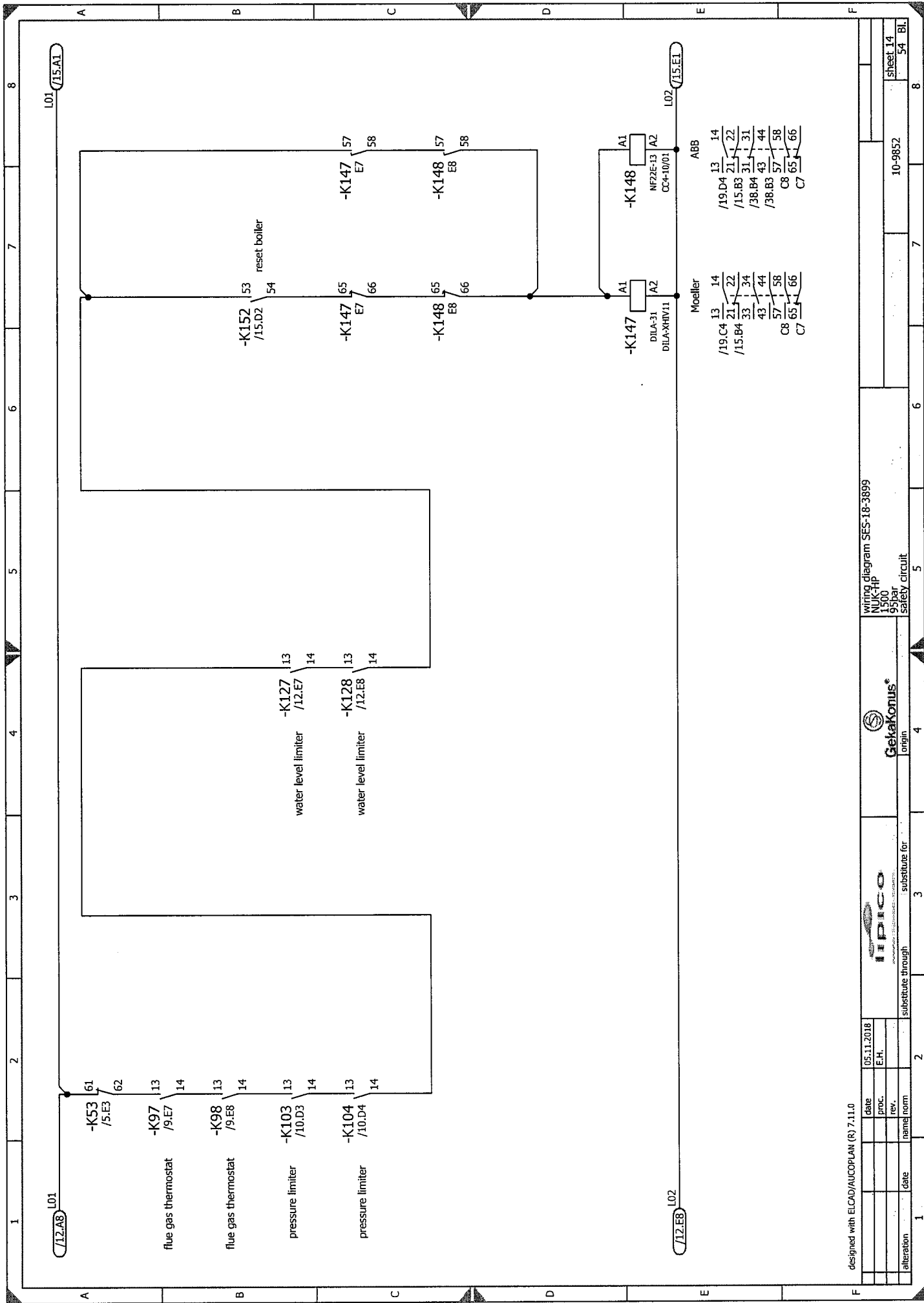
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	proc.		
	rev.		

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IPIDICO
origin

wiring diagram SES-18-3899
NOK-HP
9530
safety circuit

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sheet 13
54 BL



designed with ELCAD/AUCOPLAN (R) 7.11.0

wiring diagram SES-18-3899
NOK-HP
950
950
Safety circuit

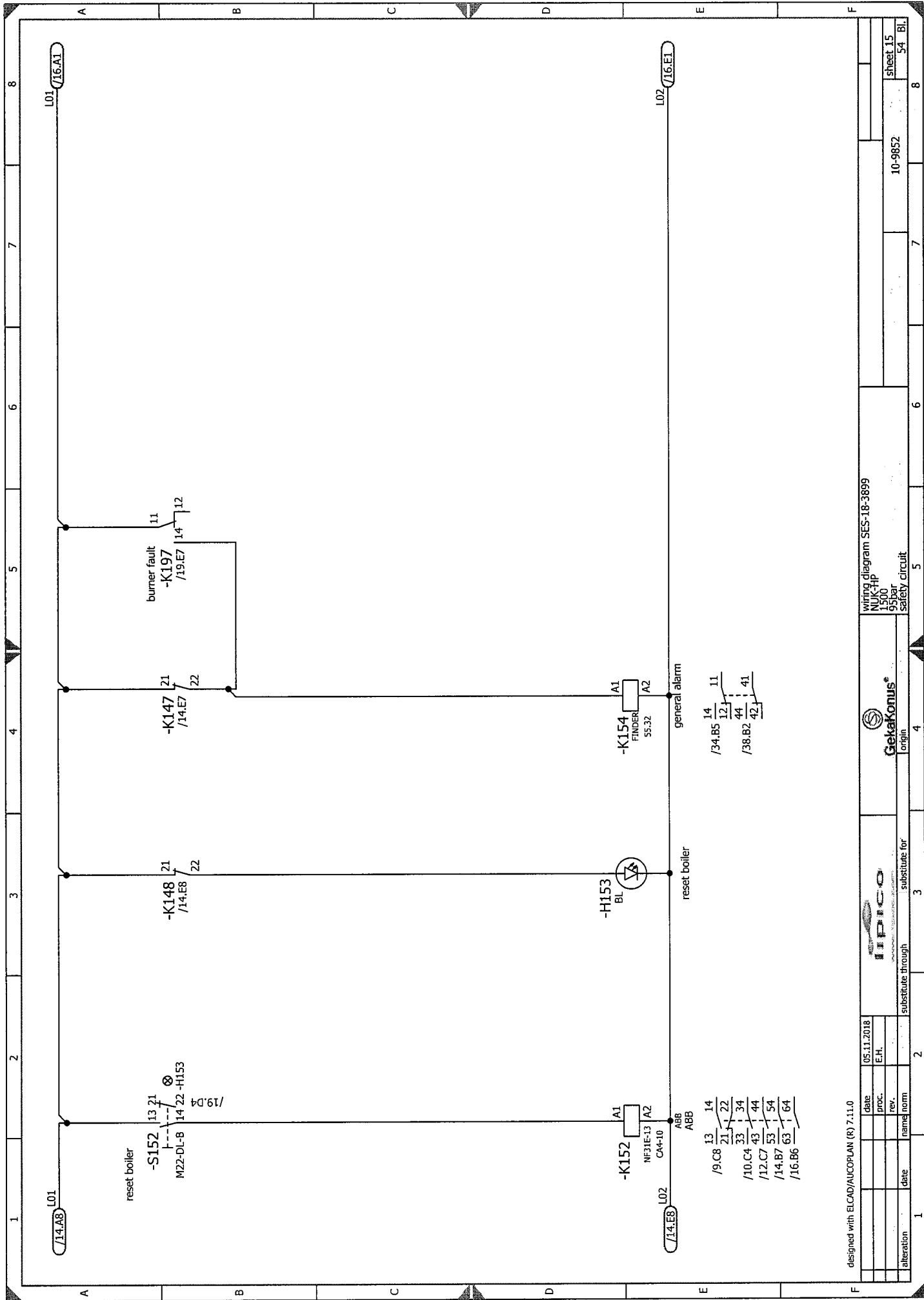
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origin

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substitute for

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					E.H.	05.11.2018

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wiring diagram SES-18-3899

NUK-HP

9500

safety circuit



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E.H.

date

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E.H.

date

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proc.

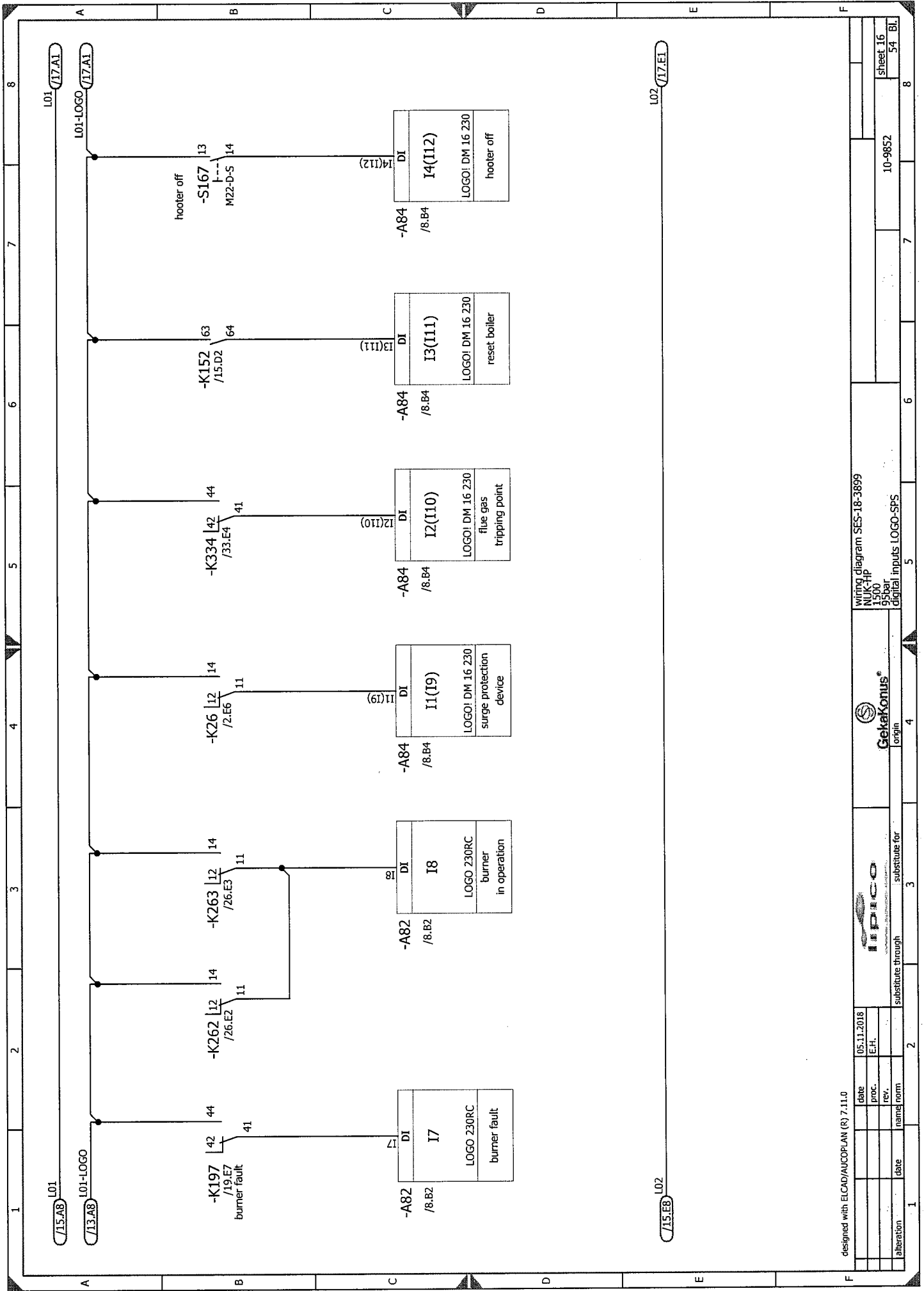
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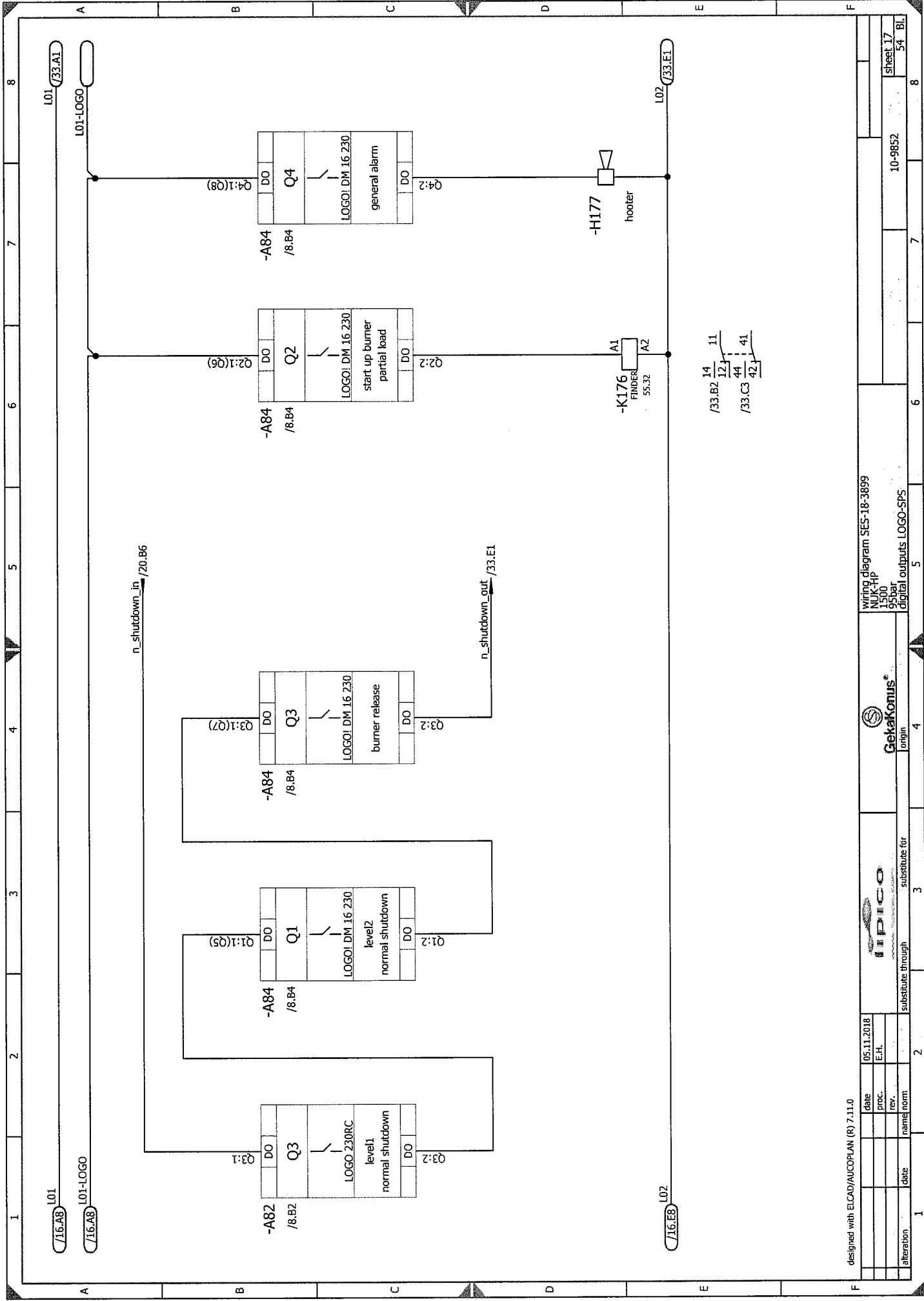
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LOGO-SPS

wiring diagram SES-18-3899
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5300
digital inputs LOGO-SPS

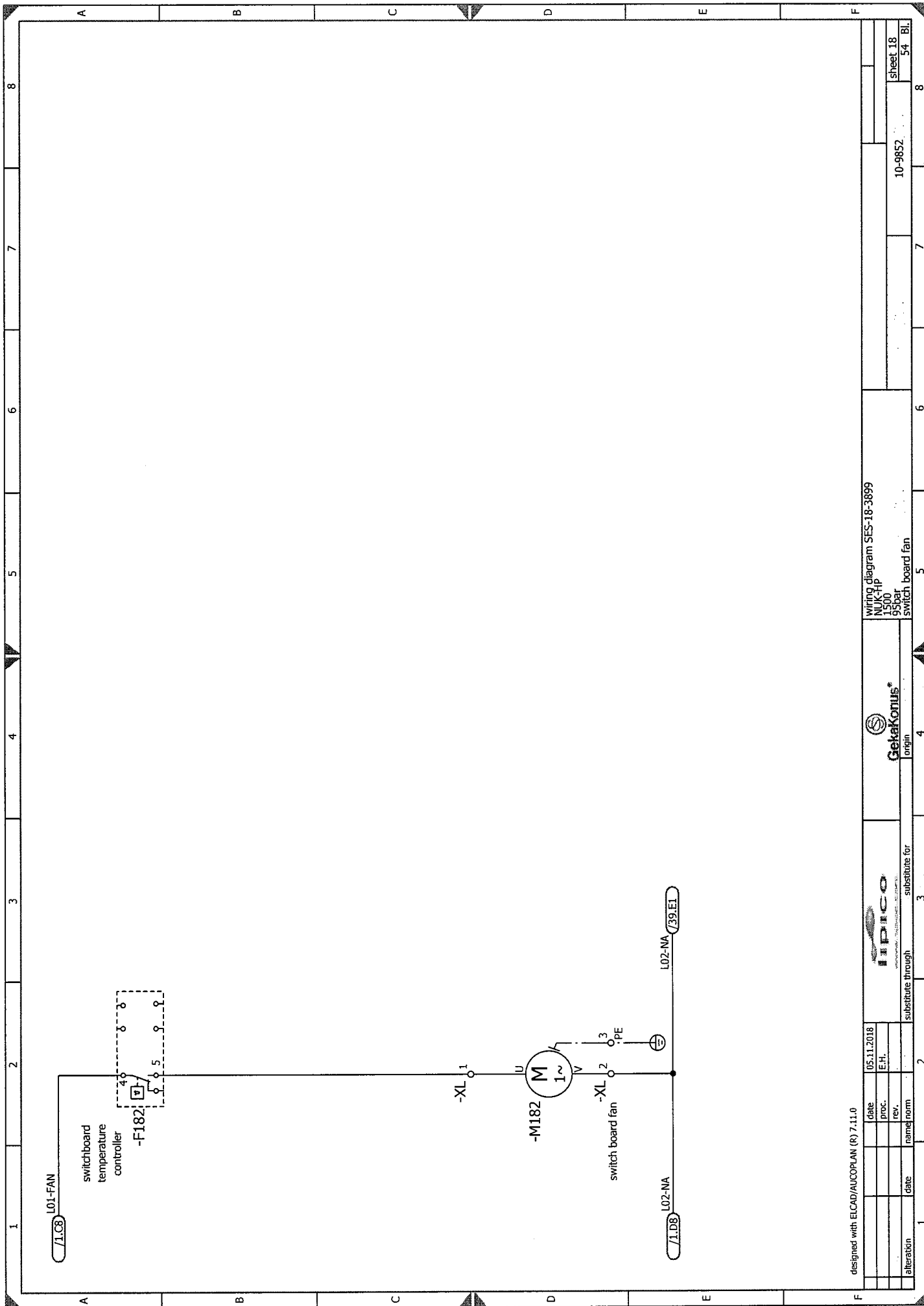
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14 11
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/33.C3 44 41 42

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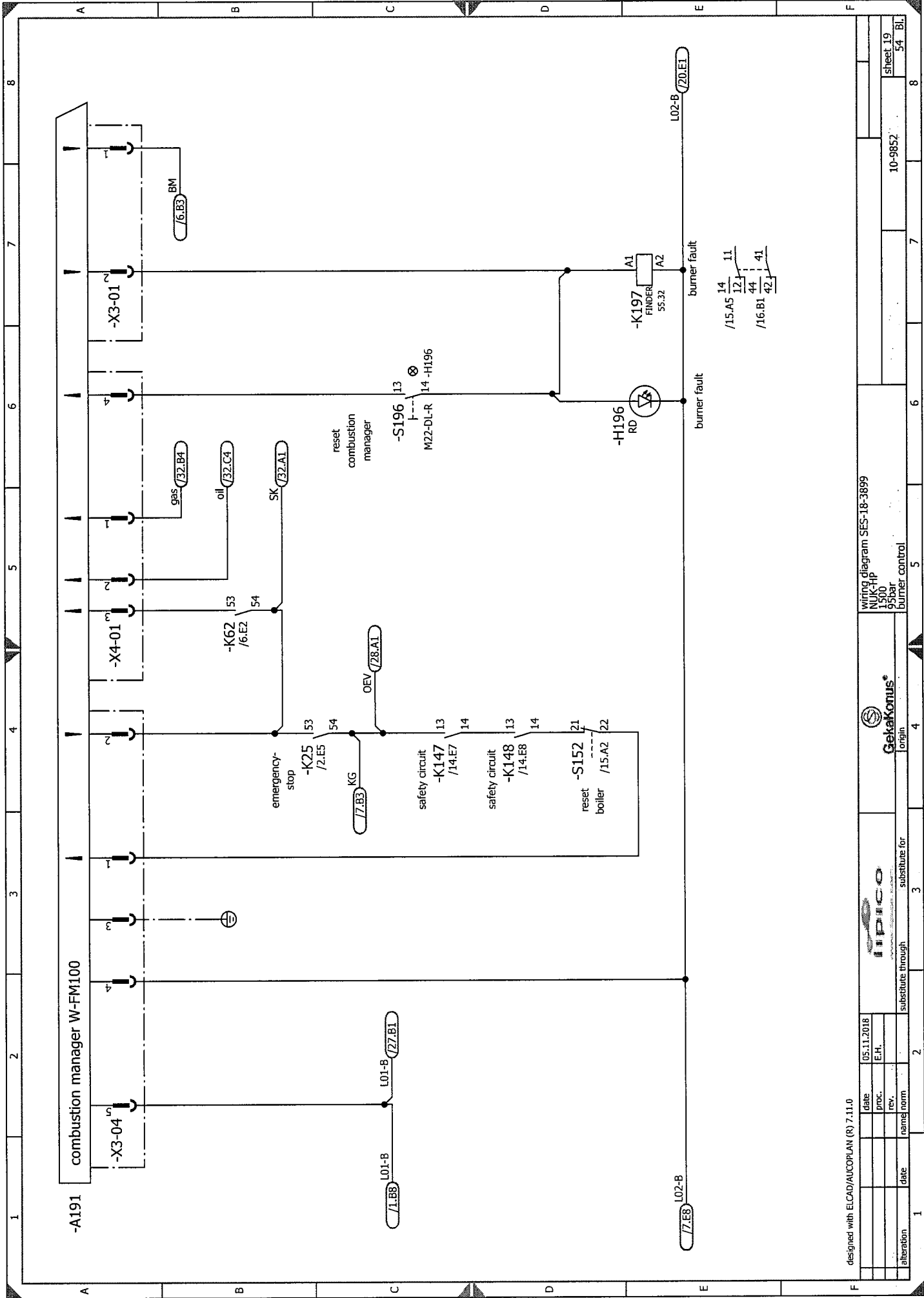
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origin

wiring diagram SES-18-3899
NKK-HP
1000
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switch board fan

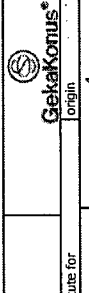
sheet 18
54 Bl.



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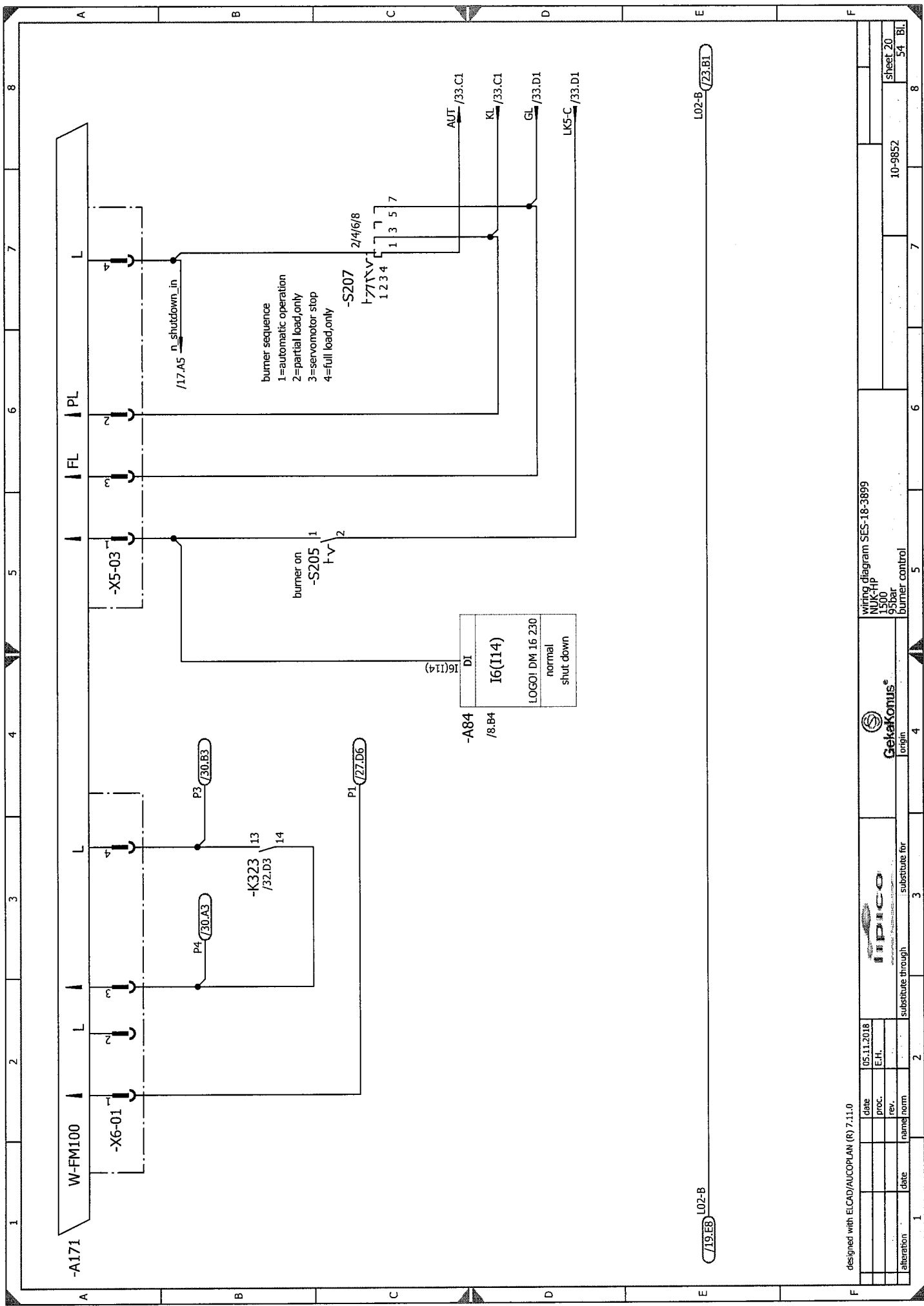
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	05.11.2018	proc.	E.H.
		rev.	

substitute through	substitute for



wiring diagram SES-18-3899
 V0.0
 9504
 burner control

sheet 19
54 BL



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alteration	date	name	norm

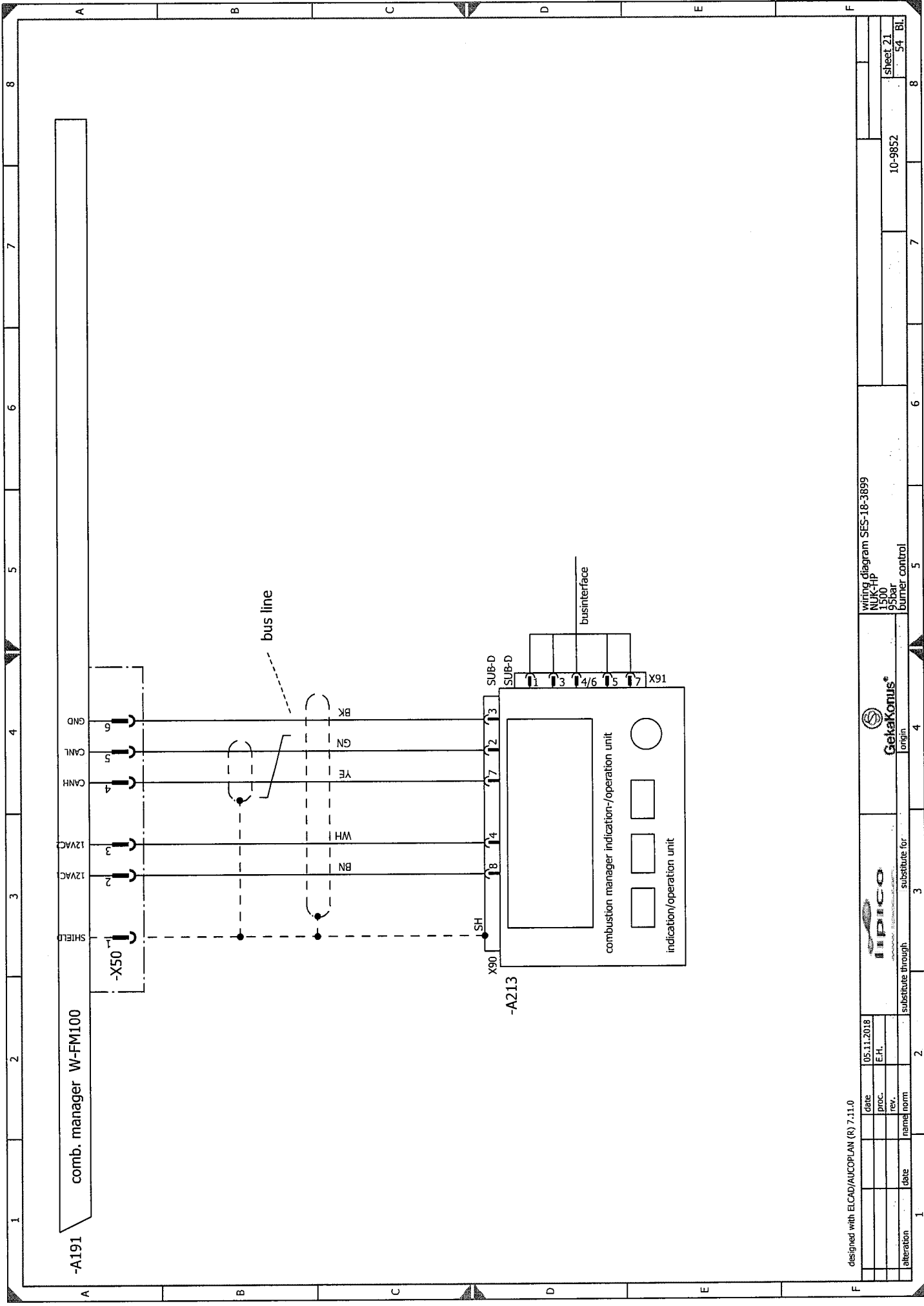
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05.11.2018	E.H.	

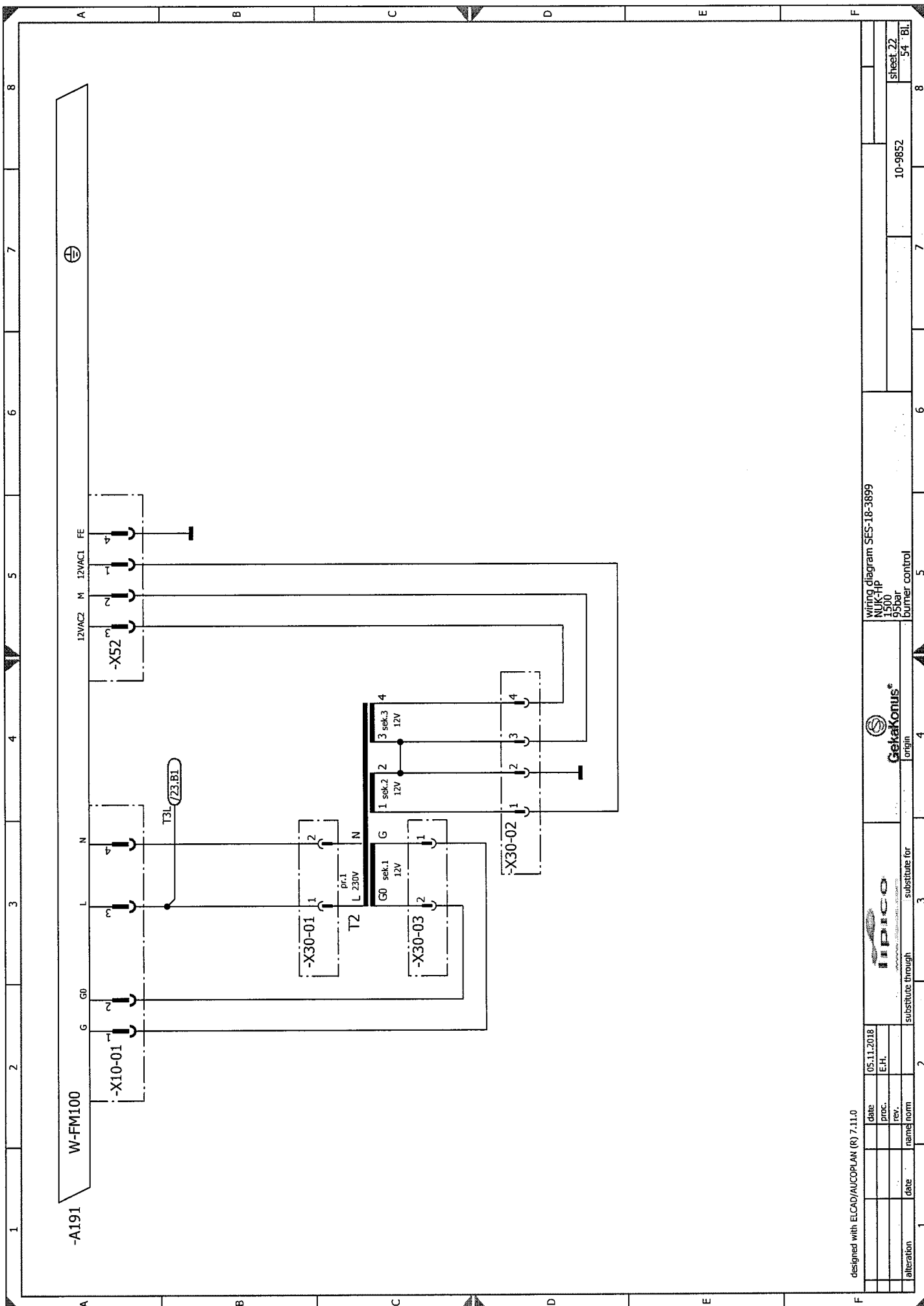
substitute through	substitute for

origin	burner control

10-9852	sheet 20
	54 BL

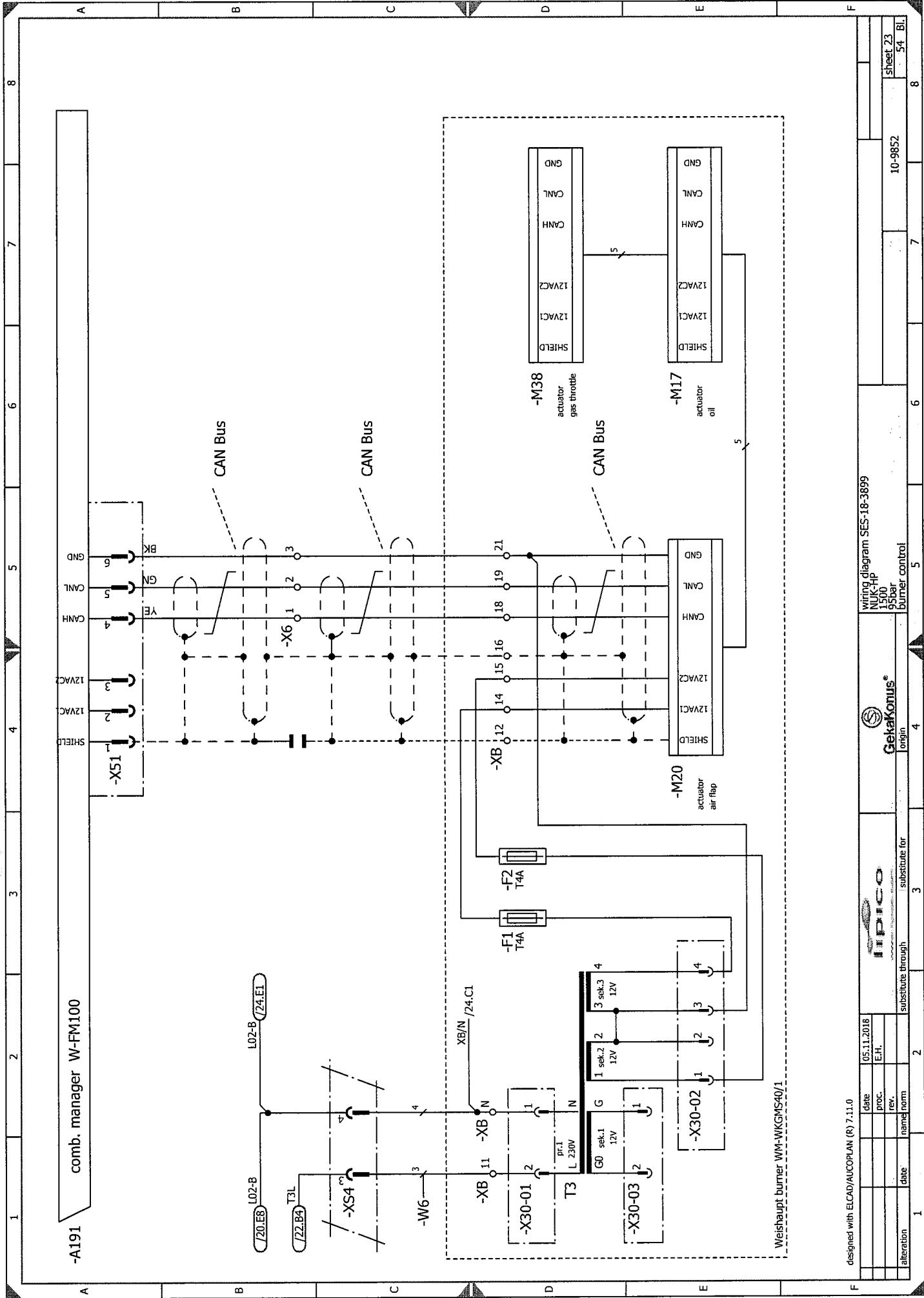
wiring diagram SES-18-3899	sheet 20
	54 BL





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alteration	date	name	norm	2	1
	05.11.2018	proc.	E.H.		
		rev.			
		substitute through		3	
		substitute for			
		origin		4	
		Gekakonus*			
		wiring diagram SES-18-3899			
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		9.5 bar			
		burner control			
		10-9852		6	7
		sheet 22			8
		54 Bl.			

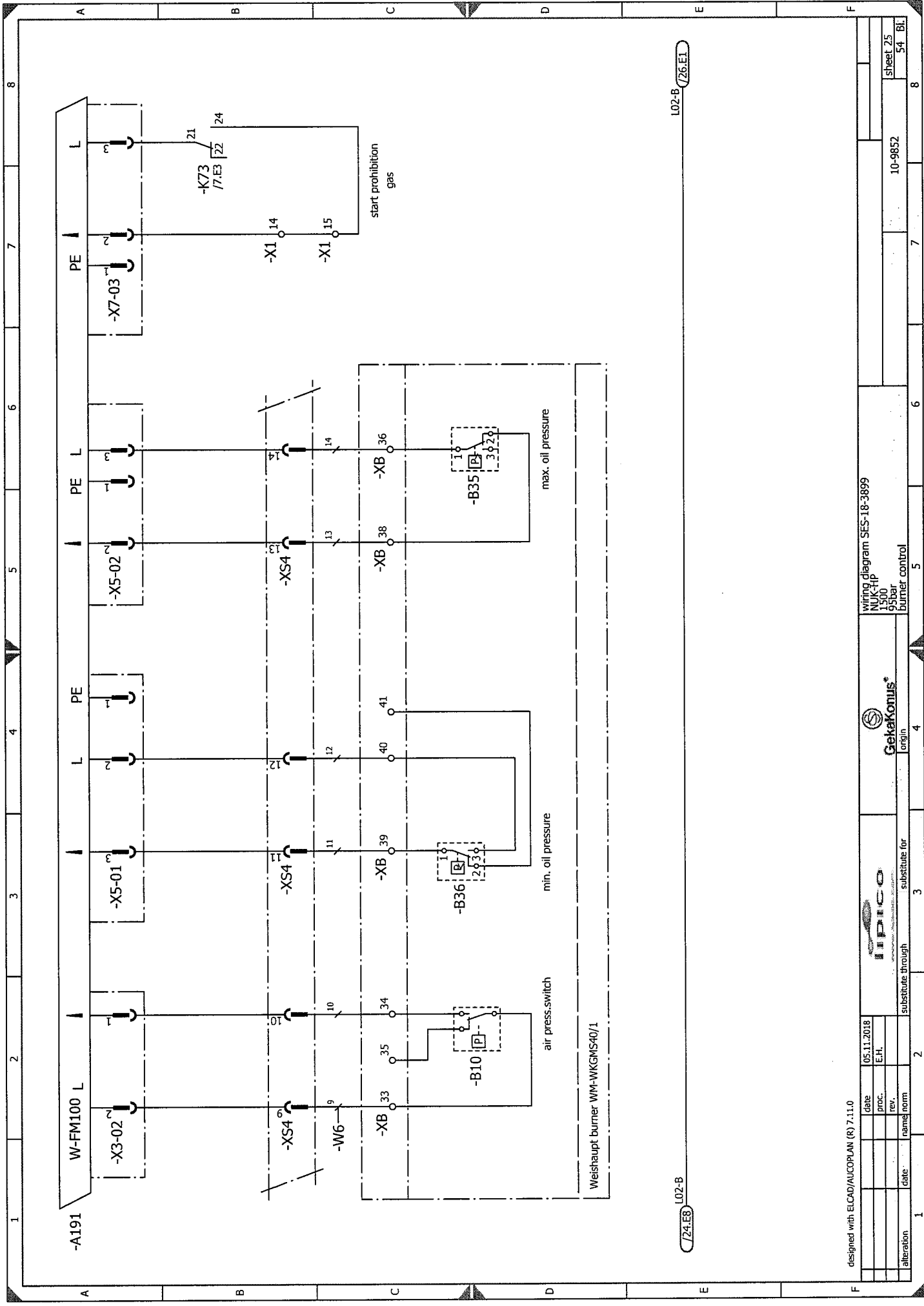


-A191 comb. manager W-FM100

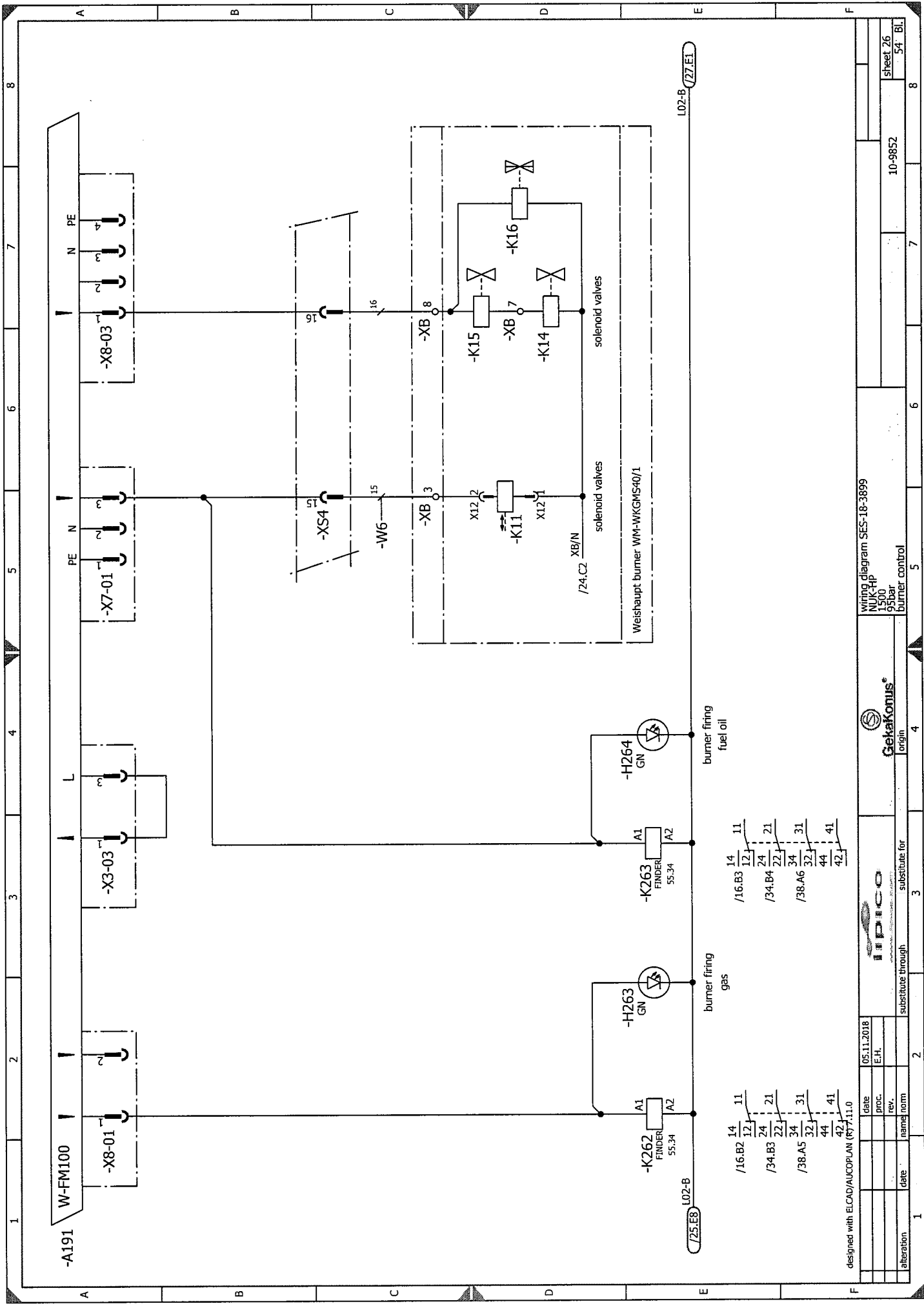
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designed with ELCAD/ALUCOPLAN (R) 7.11.0

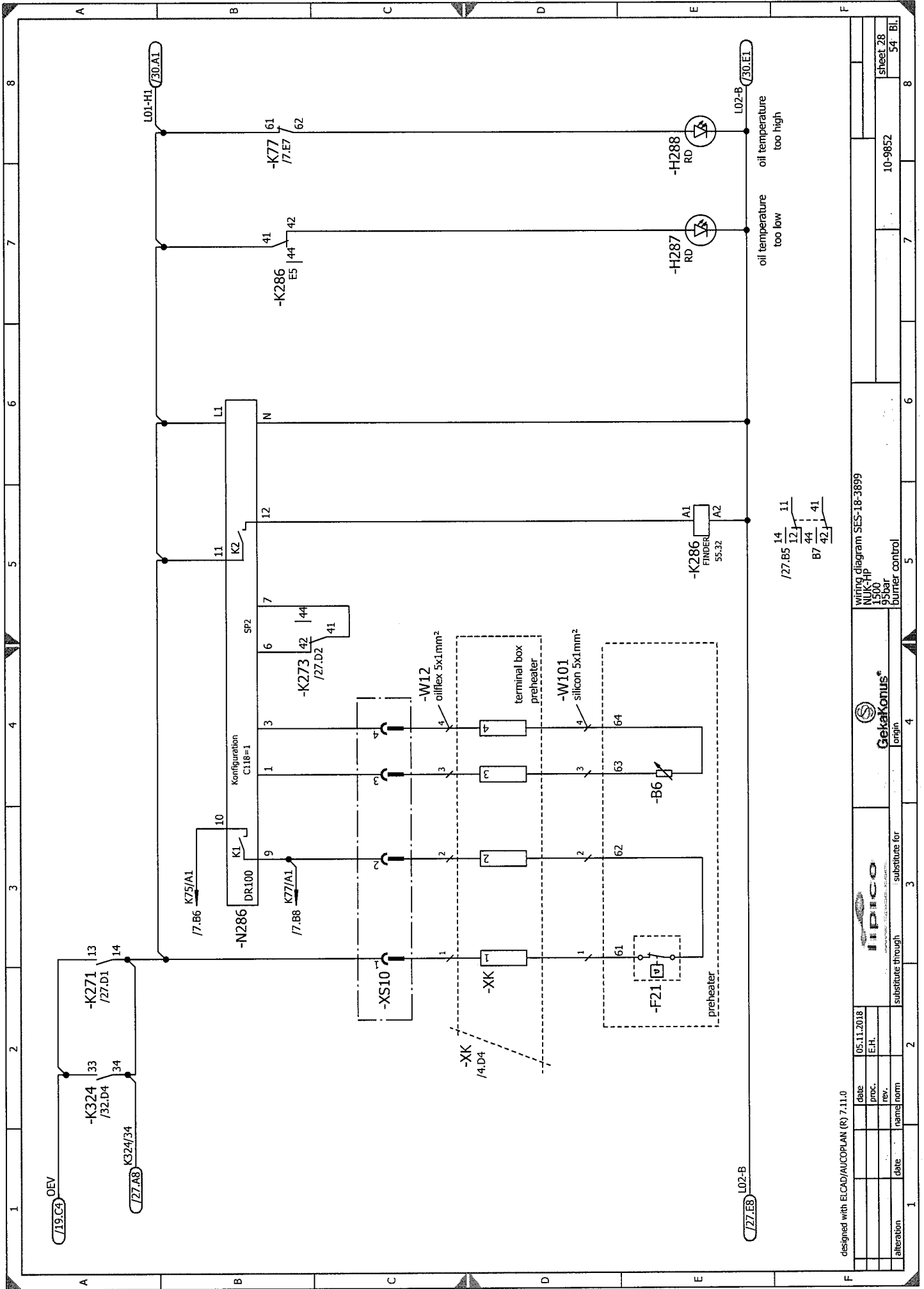
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		05.11.2018					NUK-Hp		9300							
		proc.	E.H.				burner control									



L24.EB
L02-B
L02-B
L26.E1



alteration		date	name	norm	substitute for		Gekakonus®		Wiring diagram SES-18-3899		sheet 26	
proc.		date	name	norm	substitute for		Gekakonus®		Wiring diagram SES-18-3899		sheet 26	
rev.		date	name	norm	substitute for		Gekakonus®		Wiring diagram SES-18-3899		sheet 26	
E.H.		date	name	norm	substitute for		Gekakonus®		Wiring diagram SES-18-3899		sheet 26	
10-9852		date	name	norm	substitute for		Gekakonus®		Wiring diagram SES-18-3899		sheet 26	
burner control		date	name	norm	substitute for		Gekakonus®		Wiring diagram SES-18-3899		sheet 26	
1500		date	name	norm	substitute for		Gekakonus®		Wiring diagram SES-18-3899		sheet 26	
54' BL		date	name	norm	substitute for		Gekakonus®		Wiring diagram SES-18-3899		sheet 26	



designed with ELCAD/AUCOPLAN (R) 7.11.0

alteration	date	name	nom	rev	proc.	date
						05.11.2018
						E.H.

substitute through

substitute for

burner control

10-9852

sheet 28

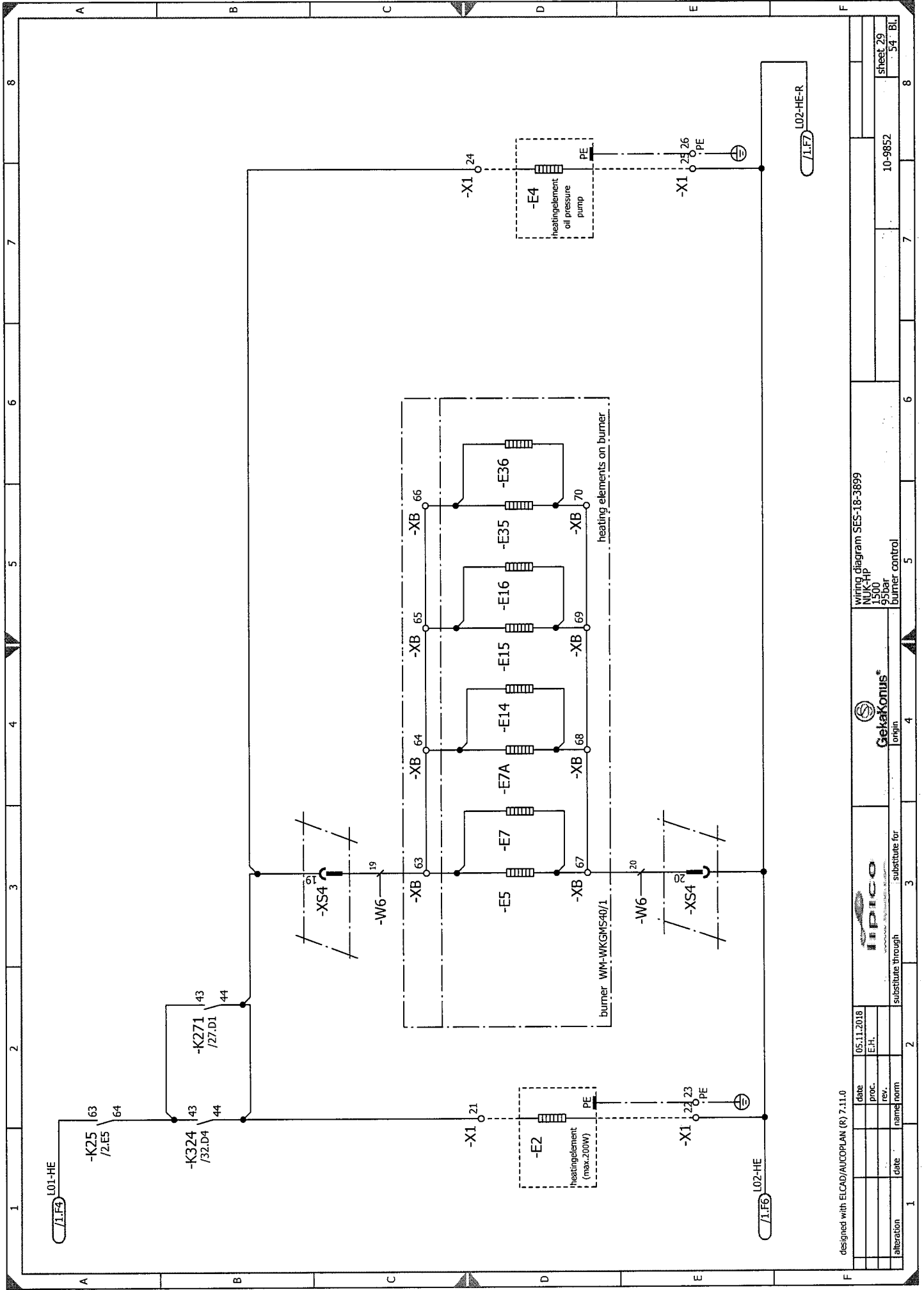
54 Bl.

10-9852

sheet 28

54 Bl.

10-9852



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wiring diagram SES-18-3899

NUK-HP

substitute for

substitute through

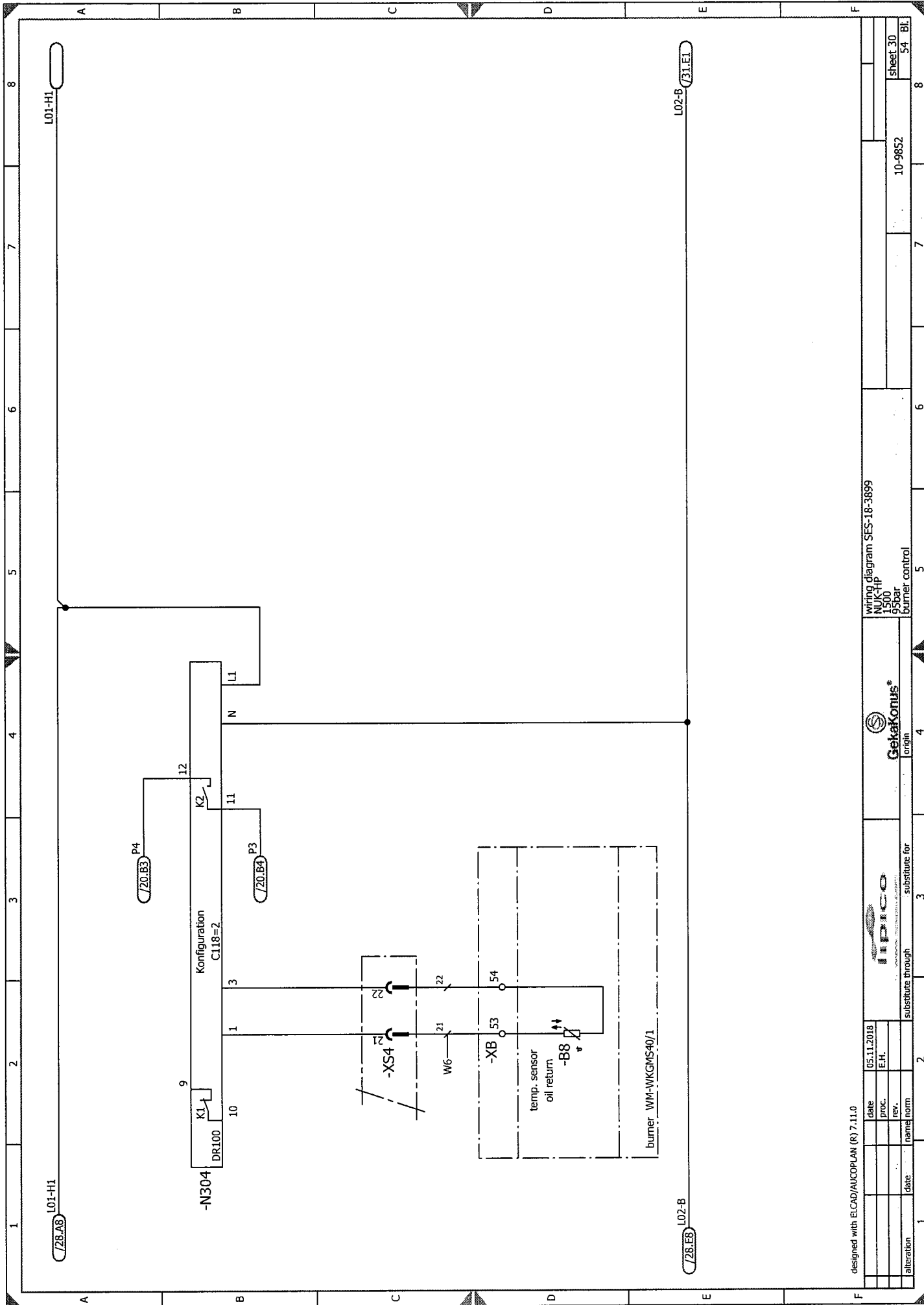
origin

Gekakonus

sheet 29

10-9852

54 Bl.



designed with ELCAD/AUCOPLAN (R) 7.11.0

date 05.11.2018

proc. E.H.

rev.

name norm

alteration

date

substitute through

substitute for

origin

Gekakonus

wiring diagram SES-18-3899

1000 HP

9.5 bar

burner control

10-9852

sheet 30

54 Bl.

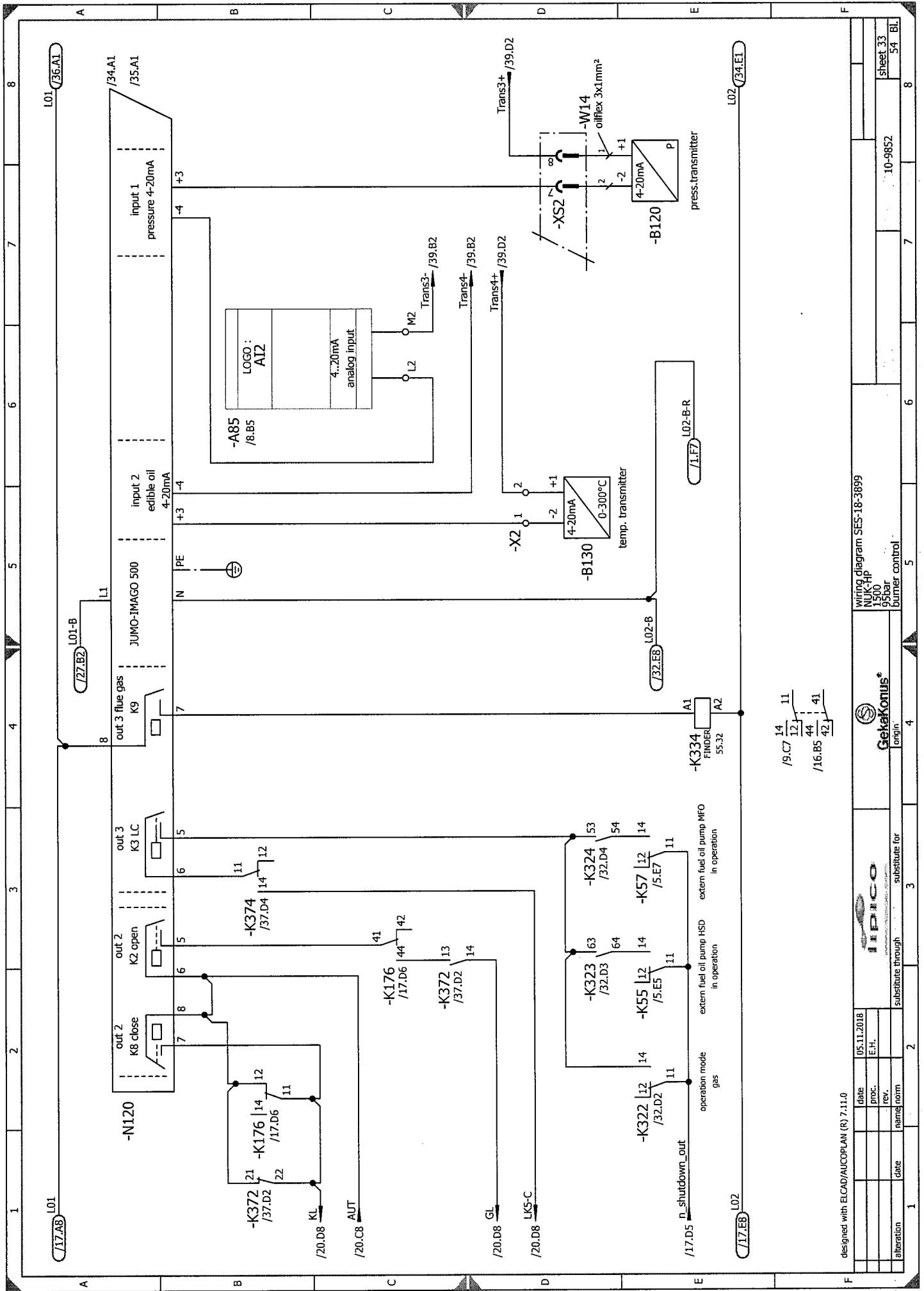
L01-H1

L02-B

L01-H1

L02-B





14 11
/9.C7 12
/16.B5 44 41
42

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alteration	date	name	form	rev.	proc.	date
					E.H.	05.11.2018

substitute through

substitute for

Gekakonus®

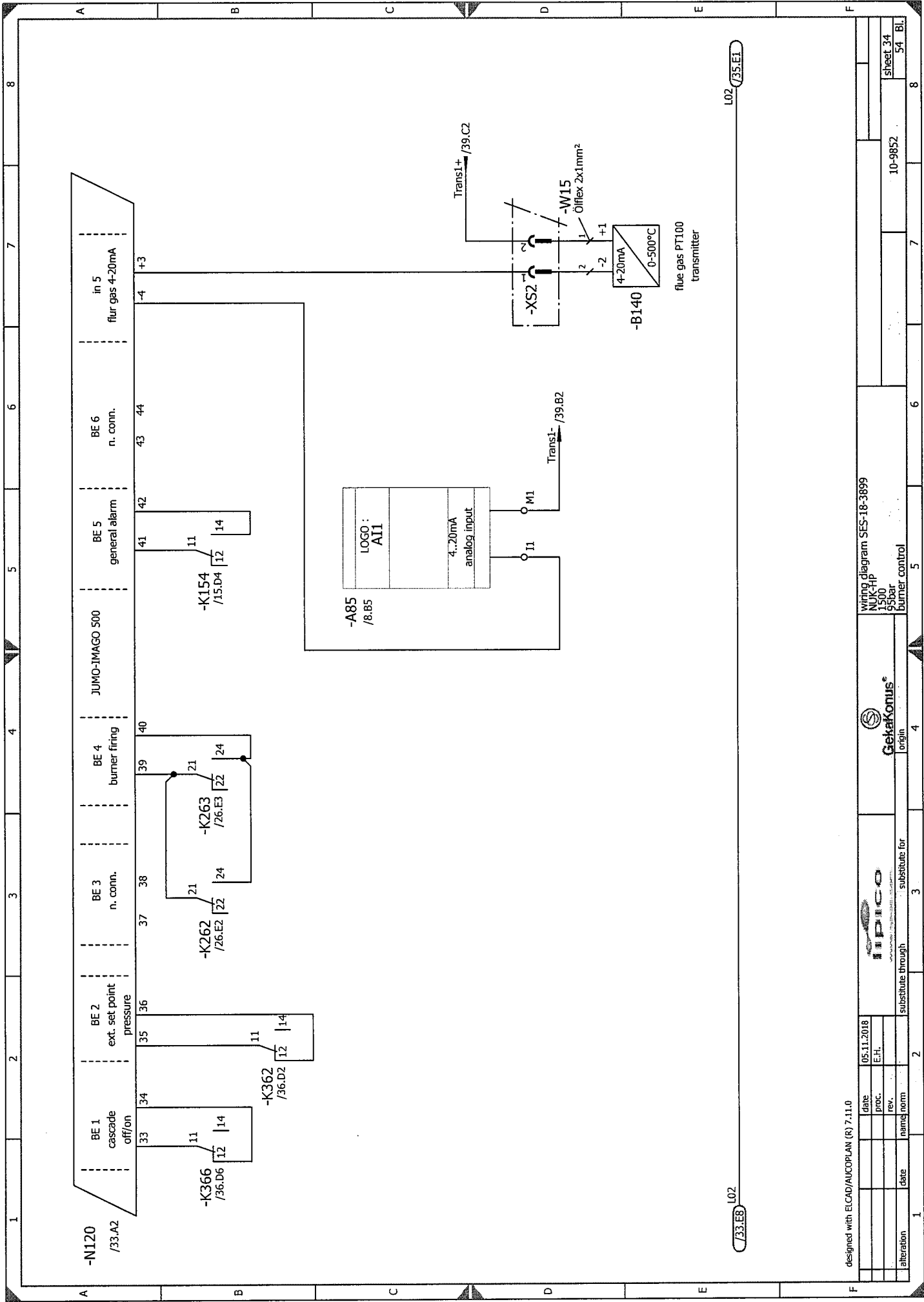
wiring diagram SES-18-3899

NUK-Hp

10-9852

sheet 33

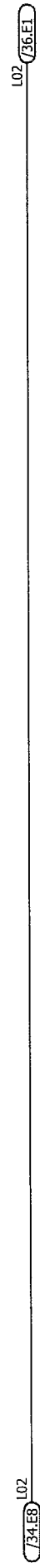
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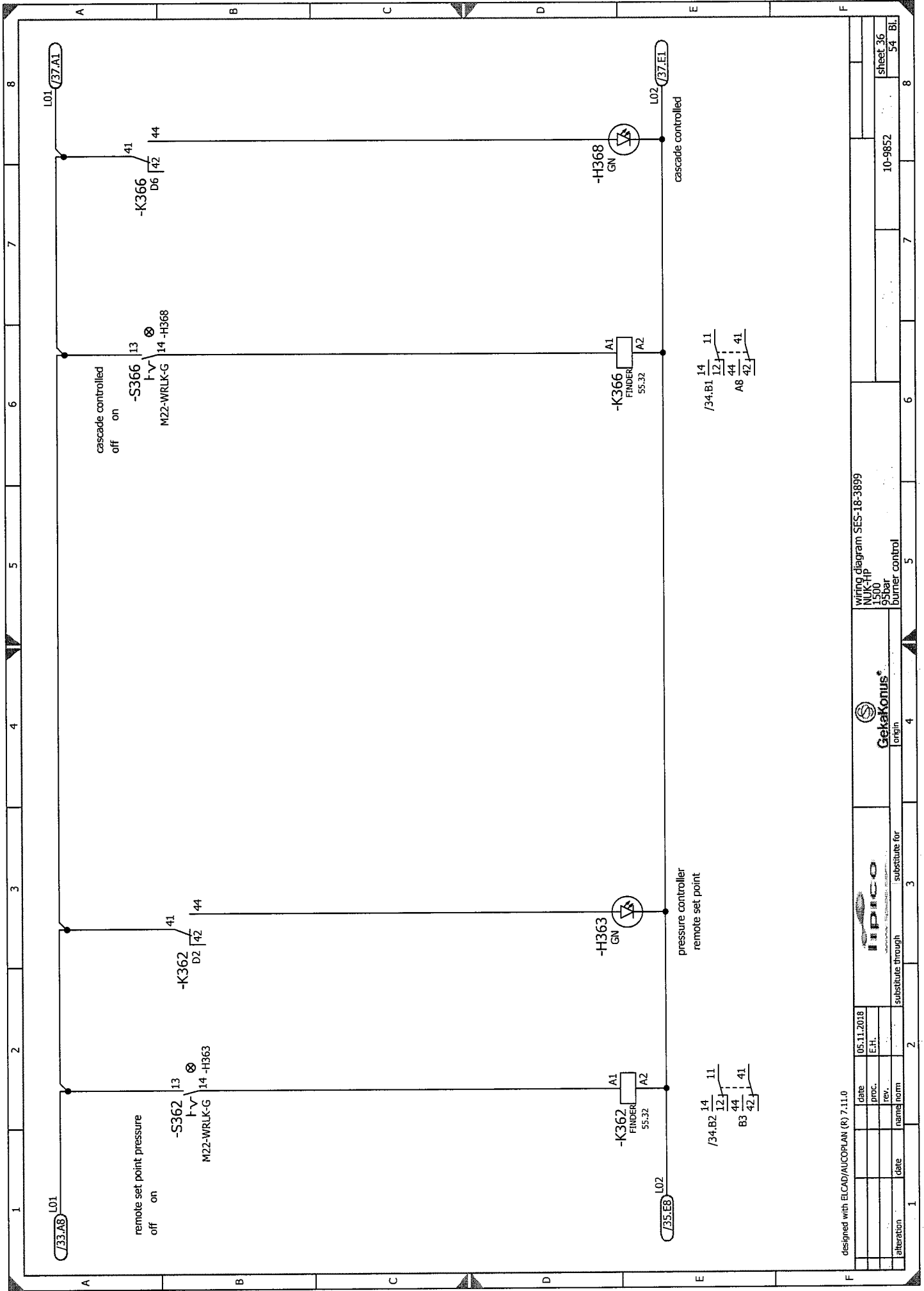


designed with ELCAD/AUCOPLAN (R) 7.11.0

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substitute through		substitute for			origin		Gekakonus*			
1		2			3		4			
10-9852		1500			1500		1500			
54 Bl.		burner control			burner control		burner control			
sheet 34		10-9852			10-9852		10-9852			
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wiring diagram SES-18-3899

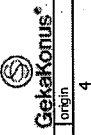




designed with ELCAD/AUCOPPLAN (R) 7.11.0

alteration	date	named form	rev.	proc.	date
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					E.H.

substitute through	substitute for



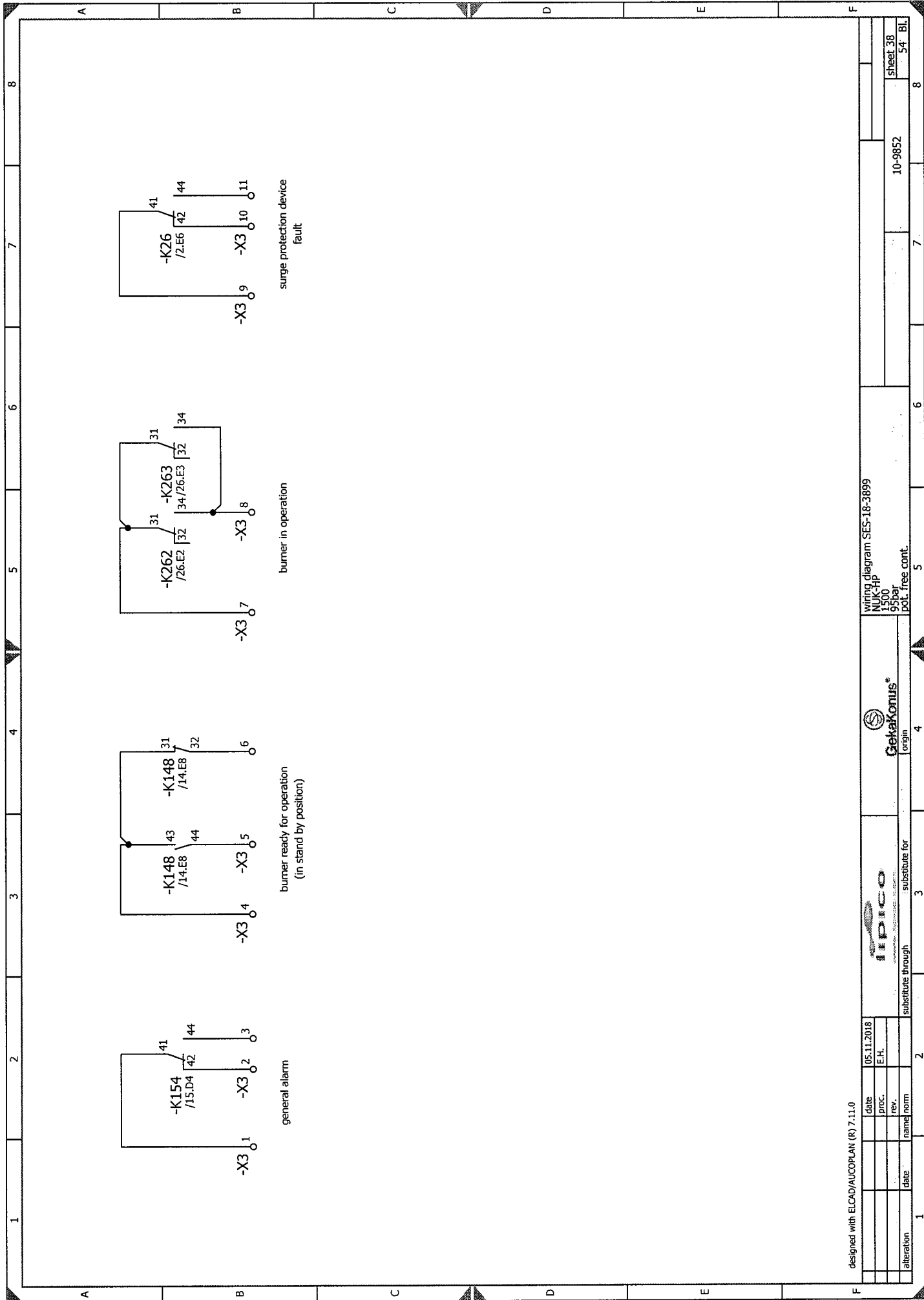
wiring diagram SES-18-3899

NUK-HP
3500
burner control

10-9852

sheet 36
54 BL





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alteration	date	name	norm	rev.	proc.	date
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substitute through	3	4
substitute for	3	4

IPICO

substitute for

wiring diagram SES-18-3899

NUK-HP

500

93par

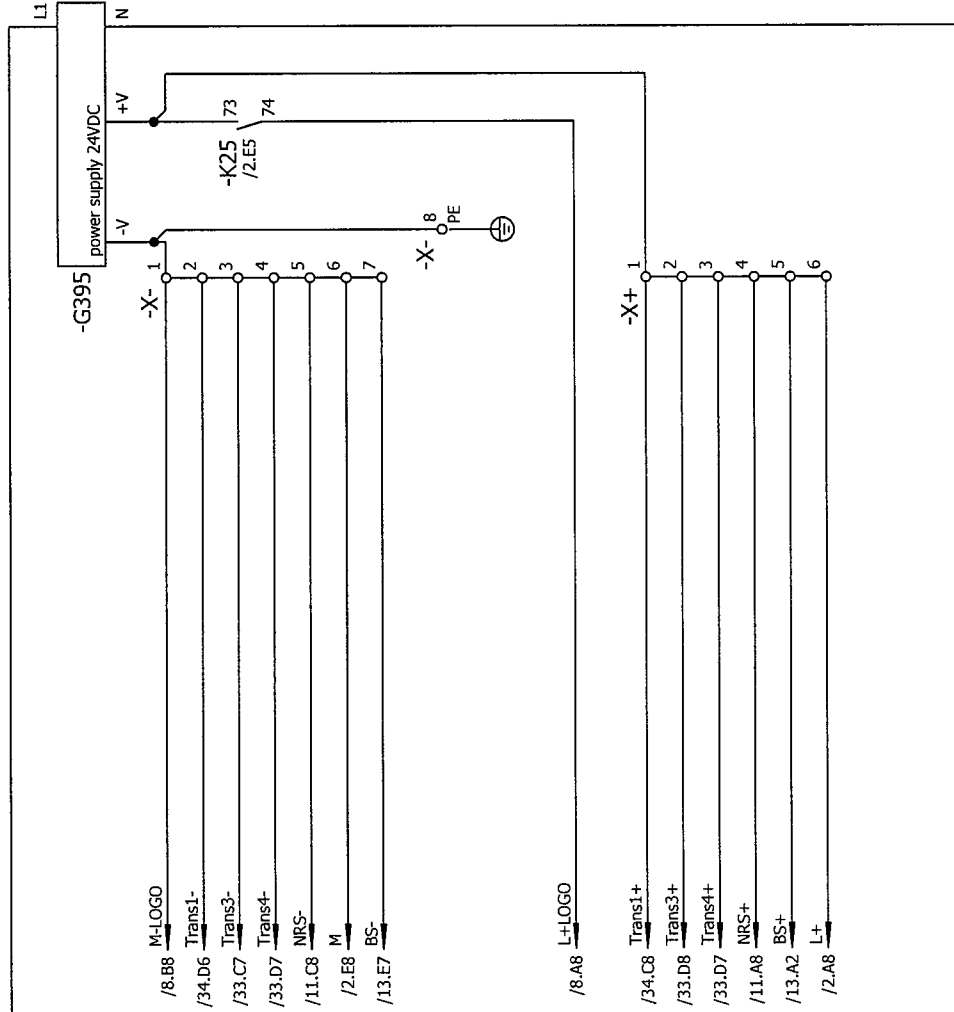
pol. free cont.

10-9852

sheet 38

54 BL

11.88 L01-NA



11.83 L02-NA

designed with ELCAD/AUCOPLAN (R) 7.11.0

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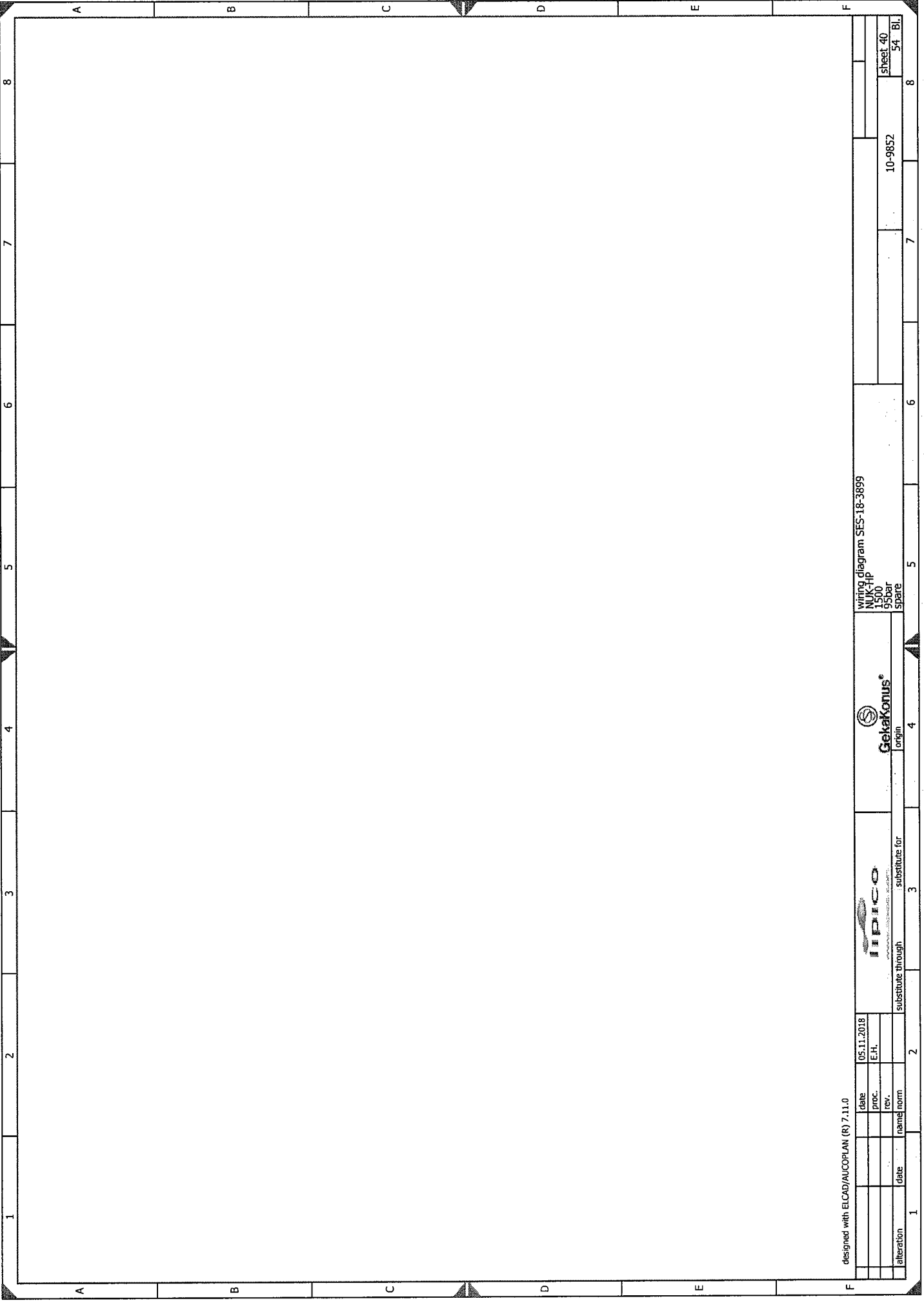
substitute through
substitute for

IPICO
Gekakonus®
origin

wiring diagram SES-18-3899
NUR-HP
1500
9500
control voltage 24VDC

10-9852

sheet 39
54 Bl.

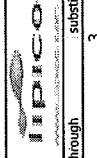


designed with ELCAD/AUCOPLAN (R) 7.11.0

alteration	date	name	norm	date	proc.	rev.	date
							05.11.2018
							E.H.

substitute through

substitute for


IPICO
INDUSTRIAL PROCESSING COMPANY


Gekakonus®
longin

wiring diagram SES-18-3899
NUK-HP
3500
95bar
spare

10-9852	sheet 40
54 Bl.	

[illegible]

designed with AUTOTEC-ELCAD (R) 7.11.0			
		burner fan	
cable mark comment	extern	-M110	U1
	target mark	-M110	U1
cable mark extern			1
			2
			3
			4
			5
cable mark			6
			7
			8
			9
			10
connection strip			11
			12
			13
			14
			15
appliance comment			16
			17
			18
			19
			20
connection strip			21
			22
			23
			24
			25
appliance comment			26
			27
			28
			29
			30
connection strip			31
			32
			33
			34
			35
appliance comment			36
			37
			38
			39
			40
connection strip			41
			42
			43
			44
			45
appliance comment			46
			47
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connection strip			51
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appliance comment			56
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connection strip			61
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appliance comment			66
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connection strip			71
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appliance comment			76
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connection strip			81
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appliance comment			86
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connection strip			91
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appliance comment			136
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connection strip			141
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[illegible]